



COM 682-Week 9

Cloud Native Development

Load Balancing

Azure Monitor & Azure Application Insights Profiler

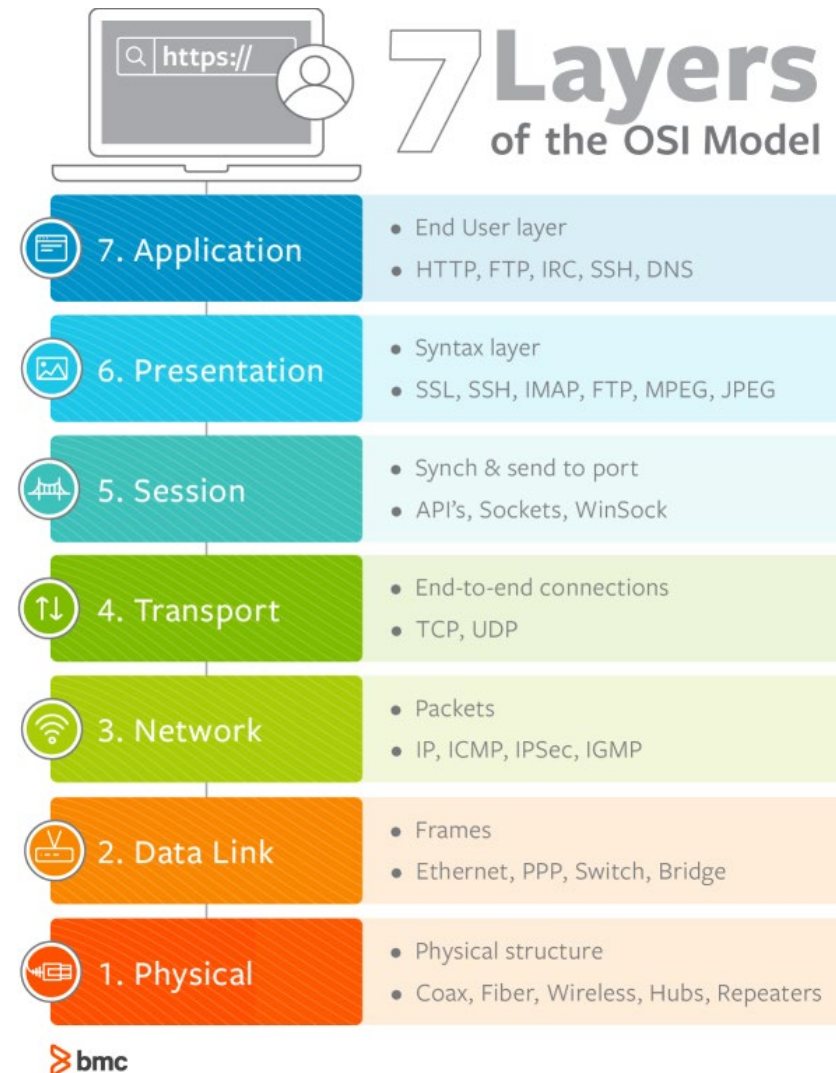
Agenda

- Load balancing
- Azure Monitor
- Azure Application Insights Profiler

Load Balancer

Overview

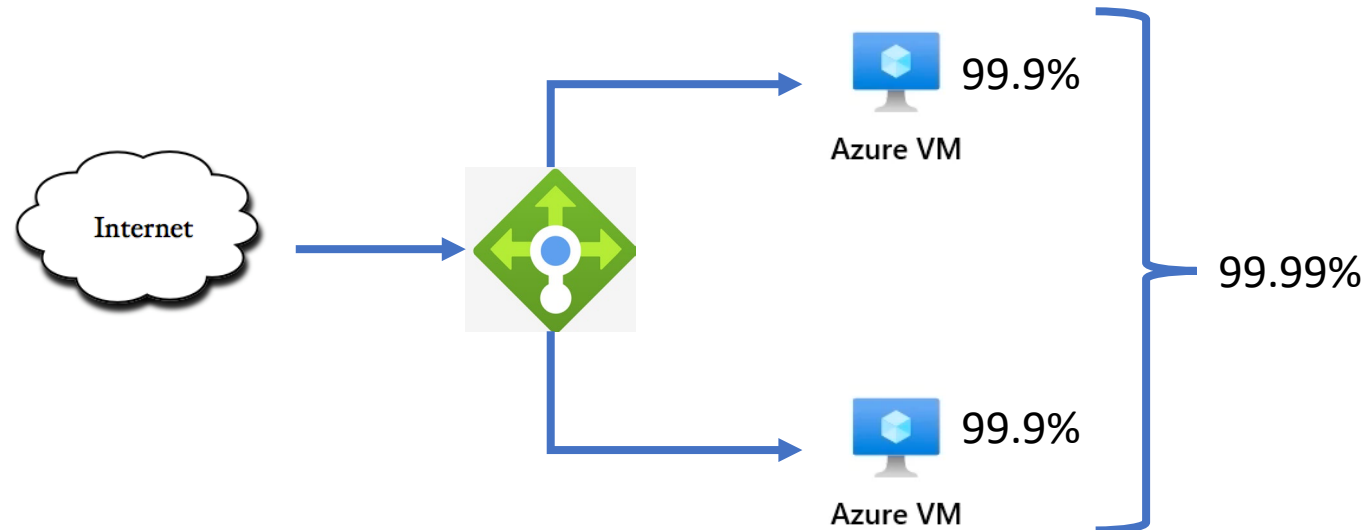
- **Azure Load Balancer operates at layer 4** of the Open Systems Interconnection (OSI) model.
- **Load balancer distributes inbound flows** that arrive at the load balancer's front end to backend pool instances.
- These flows are according to configured load-balancing rules.
- The backend pool instances can be Azure Virtual Machines or instances in a virtual machine scale set.



Load Balancer

Why?

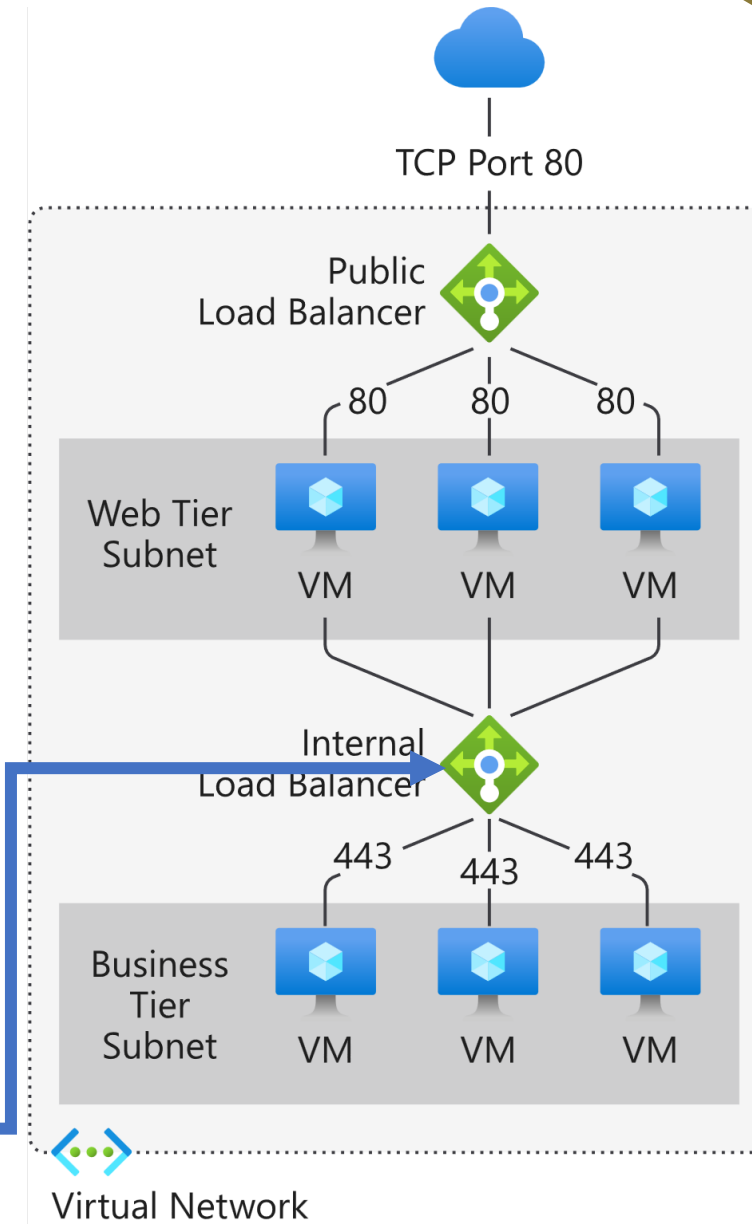
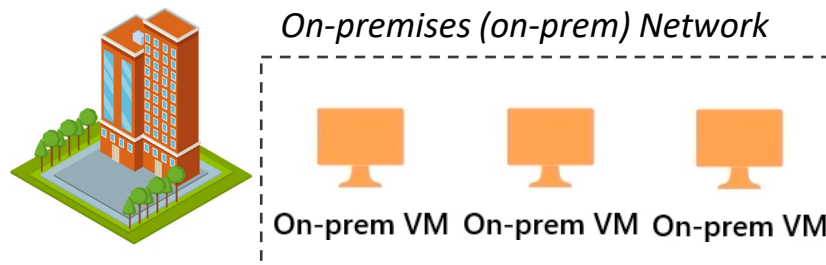
- By adding additional instances you are scaling out, which means you can accommodate much larger workloads, versus scaling up, where you simply add more power to the VM.
- By default, Azure guarantees 99.9% of network availability, but if you put those in two separate availability zones, and put a Load Balancer in front, then Microsoft guarantees 99.99% of availability. This is because the Load Balancer automatically checks the availability of the VM, and if there is any problem, it automatically redirects the traffic to the other available VM.



Load Balancer

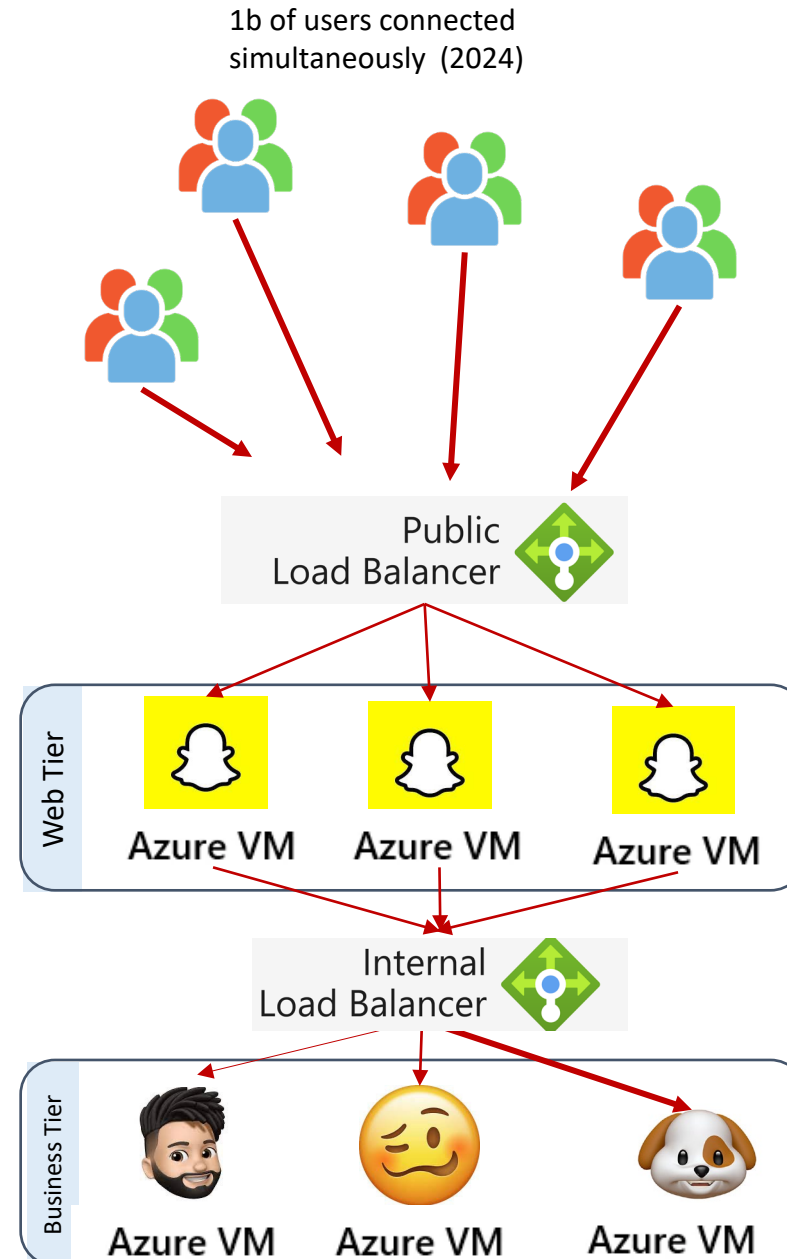
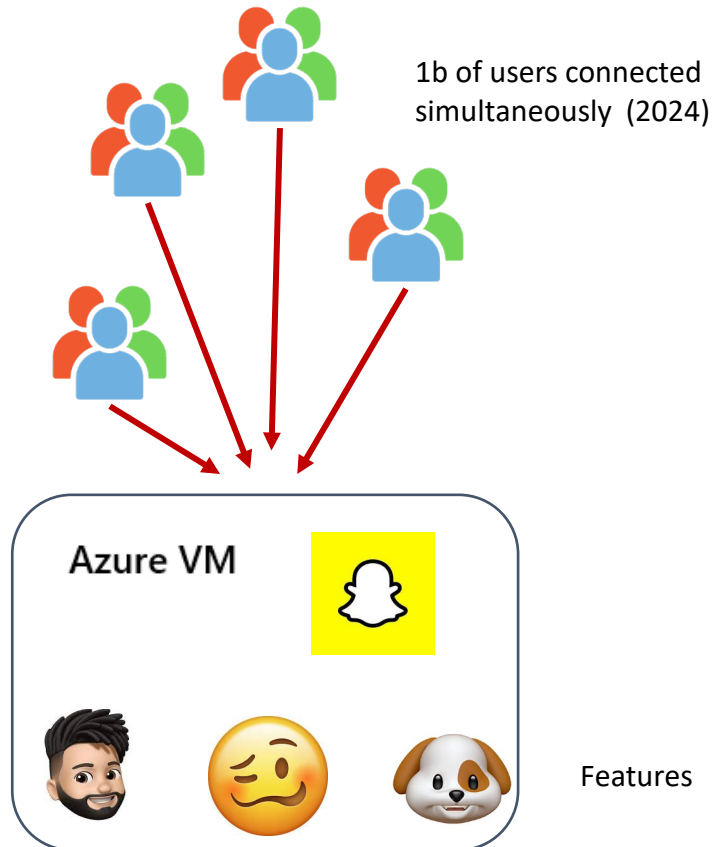
Why? – Larger Solutions

- A **public load balancer** can provide outbound connections for virtual machines (VMs) inside your virtual network. These connections are accomplished by translating their private IP addresses to public IP addresses. **Public Load Balancers are used to load balance internet traffic to your VMs.**
- An **internal (or private) load balancer** is used where private IPs are needed at the frontend only. **Internal load balancers are used to load balance traffic inside a virtual network. A load balancer frontend can be accessed from an on-premises network in a hybrid scenario.**



Load Balancer

Why? – Larger Solutions - Example



Load Balancer

Scale and Create Highly Available Services

With Azure Load Balancer, you can scale your applications and create highly available services. Load balancer supports both inbound and outbound scenarios. **Load balancer provides low latency and scales up to millions of flows for all TCP and UDP applications.**

Key scenarios that you can accomplish using Azure Standard Load Balancer include:

- **Load balance internal and external traffic** to Azure virtual machines.
- **Increase availability** by distributing resources within and across zones.
- Load balance services on **multiple ports, multiple IP addresses**, or both.
- Move internal and external **load balancer resources across Azure regions**.
- **Load balance TCP and UDP** flow on all ports simultaneously.
- Use **health probes** to monitor load-balanced resources.
- Employ **port forwarding** to access virtual machines in a virtual network by public IP address and port.
- Enable support for load balancing of **IPv6**.



Load balancing and content delivery | Load balancers

Preview

[Analyze alerts across load balancers](#)[Inspect performance of Load Balancers](#)[Review resiliency of load balancers](#)

Overview



Load balancing



Application gateways



Load balancers



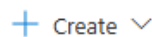
> Content delivery



> DNS load balancing



> Related services



Create



Manage view



Refresh



Export to CSV



Open query



Assign tags

[You are viewing a new version of Browse experience. Click here to access the old experience.](#)

Filter for any field...

Subscription equals all

Resource Group equals all



Location equals all



+ Add filter



No load balancers to display

Azure Load Balancer enables your applications to be highly available and scalable. You can scale up and down based on your traffic patterns. Azure Load Balancer is best suited for network traffic requiring high performance and ultra-low latency.

+ Create



Standard Load balancer

Distribute traffic to backend resources



Gateway Load balancer

Direct traffic to network virtual appliances

[Home](#) > [Load balancing and content delivery](#) | [Load balancers](#) >

Create load balancer ...

Basics

[Frontend IP configuration](#)[Backend pools](#)[Inbound rules](#)[Outbound rules](#)[Tags](#)[Review + create](#)

Azure load balancer is a layer 4 load balancer that distributes incoming traffic among healthy virtual machine instances. Load balancers use a hash-based distribution algorithm. By default, it uses a 5-tuple (source IP, source port, destination IP, destination port, protocol type) hash to map traffic to available servers. Load balancers can either be internet-facing where it is accessible via public IP addresses, or internal where it is only accessible from a virtual network. Azure load balancers also support Network Address Translation (NAT) to route traffic between public and private IP addresses. [Learn more.](#)

Project details

Subscription *

Azure for Students



Resource group *

cosmosresource

[Create new](#)

Instance details

Name *

TestLoadBlancer

Region *

Norway East

SKU * ⓘ

- ☒ Standard (Distribute traffic to backend resources)
- ☐ Gateway (Direct traffic to network virtual appliances)

Type * ⓘ

- ☐ Public
- ☒ Internal

Tier *

- ☒ Regional
- ☐ Global

Optimize every application with Azure load balancing services



[Azure Traffic Manager](#)

DNS-based traffic load balancer

Distribute traffic optimally to services across global Azure regions—with DNS-based traffic routing methods. Prioritize user access, help ensure adherence to data sovereignty, and adjust traffic for app upgrades and maintenance.

Supports:

- HTTP, HTTPS, HTTP/2, TCP, UDP
- Layer 7
- Global apps



[Azure Load Balancer](#)

Network-layer load balancer

Build high availability and network performance into your applications with low-latency layer 4 load balancing capabilities. Balance traffic between virtual machines (VMs) inside your virtual networks and across multitiered hybrid apps.

Supports:

- TCP, UDP
- Layer 4
- Global and regional apps



[Azure Application Gateway](#)

Application delivery controller as a service

Transform web front ends into scalable and highly available apps using layer 7 load balancing capabilities. Distribute regional applications more securely.

Supports:

- HTTP, HTTPS, HTTP/2
- Layer 7
- Regional apps
- Web application firewall
- SSL/TLS offloading



[Azure Front Door](#)

Highly secure global application delivery

Deliver real-time performance of global web applications using the Microsoft global edge network. Accelerate content and transform multiple microservice applications into a single more secure app delivery architecture.

Supports:

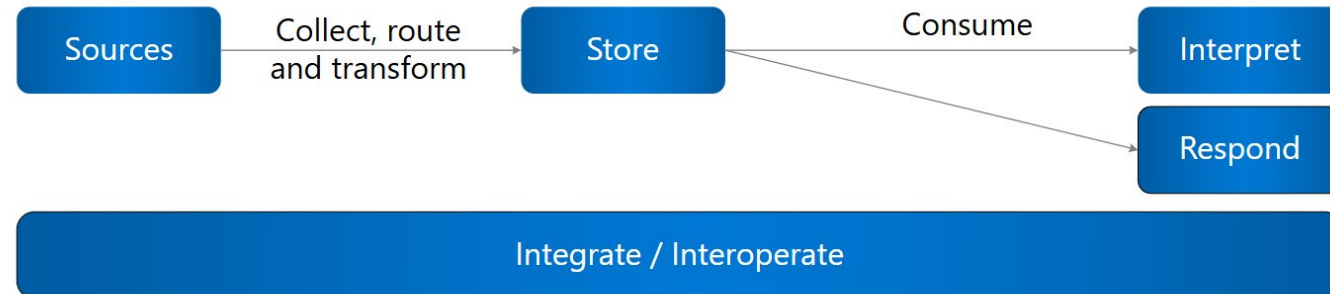
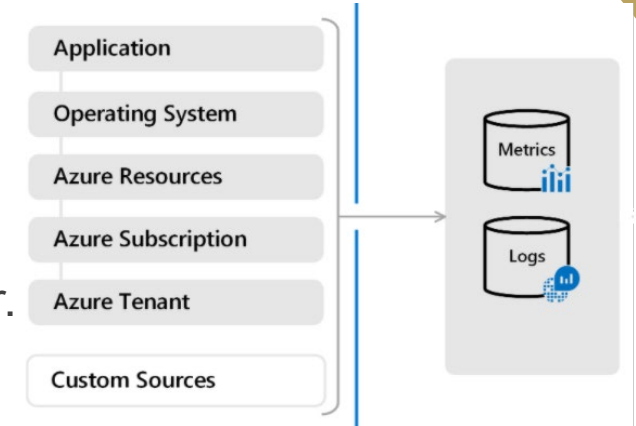
- HTTP, HTTPS, HTTP/2
- Layer 7
- Global apps
- Web application firewall
- SSL/TLS offloading



Azure Monitoring

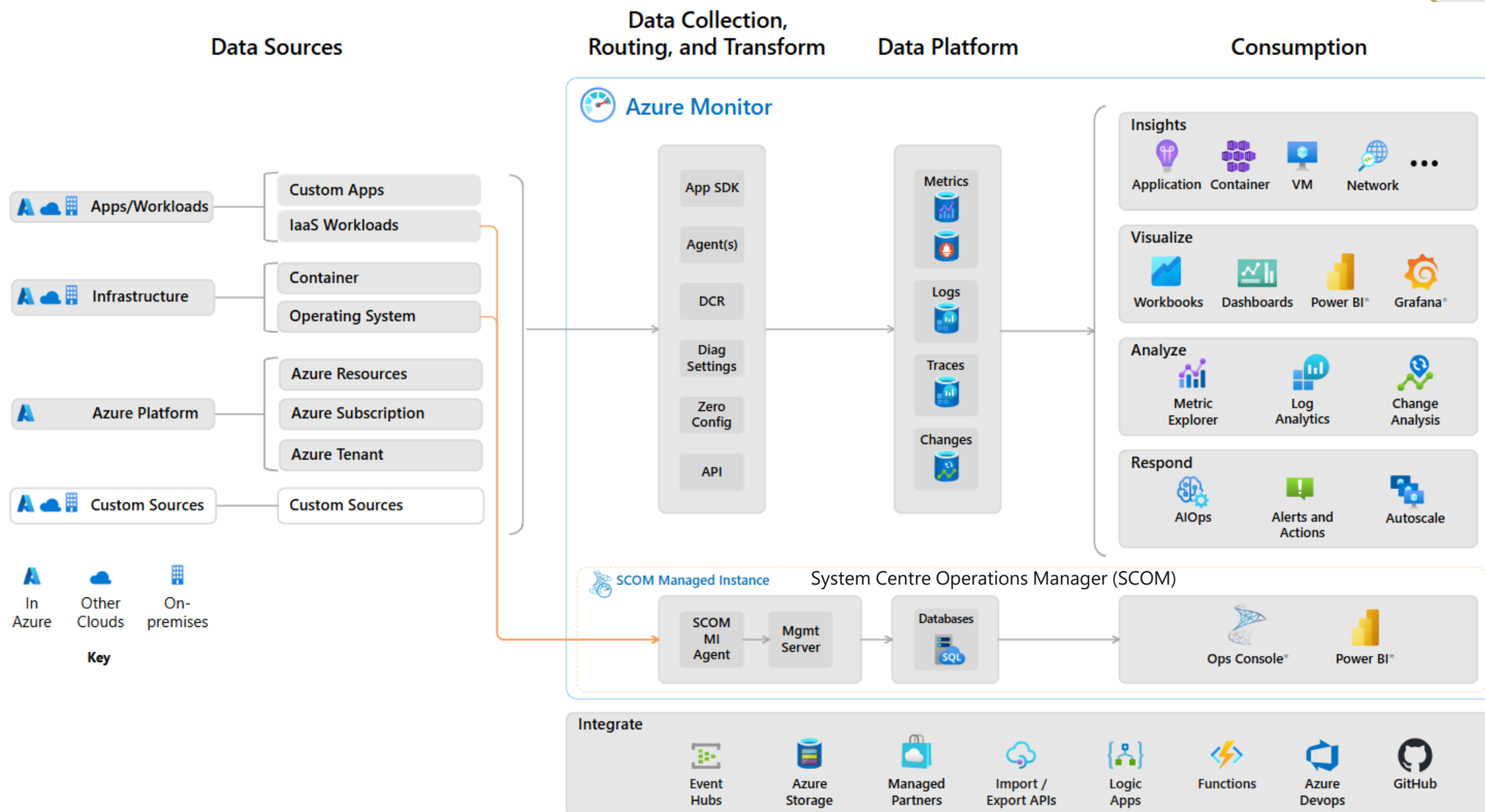
What is Monitoring?

- **Monitoring is a task that involves collecting and reviewing data about the activity of some process** or behaviour. In the world of Azure, monitoring is used to collect insights into the operation, performance, and behaviour of your applications and services.
- You can perform monitoring as **a near real-time event** or review the data later. Metrics can also be reviewed later as they are **stored for 93 days by default**. Whatever your preferred method, Azure Monitor is ready to help you collect the necessary information that you need.



Azure Monitoring

What is Monitoring?

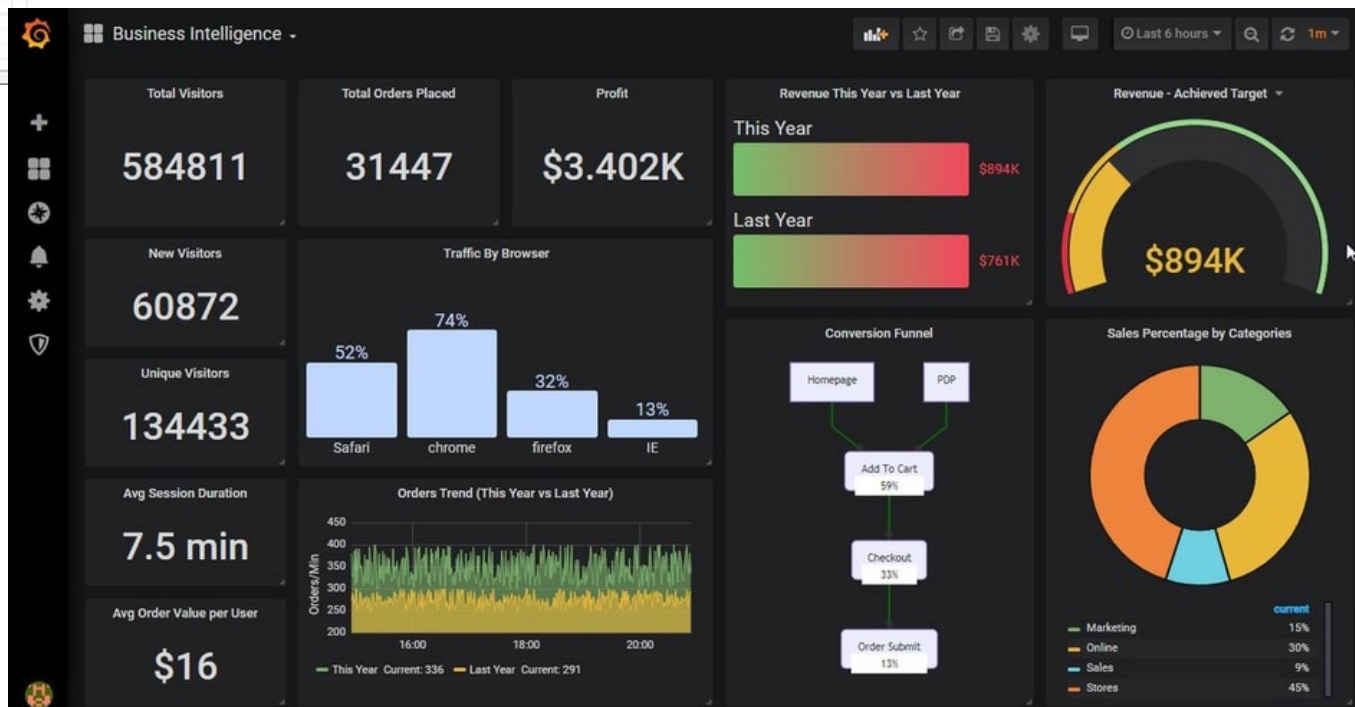
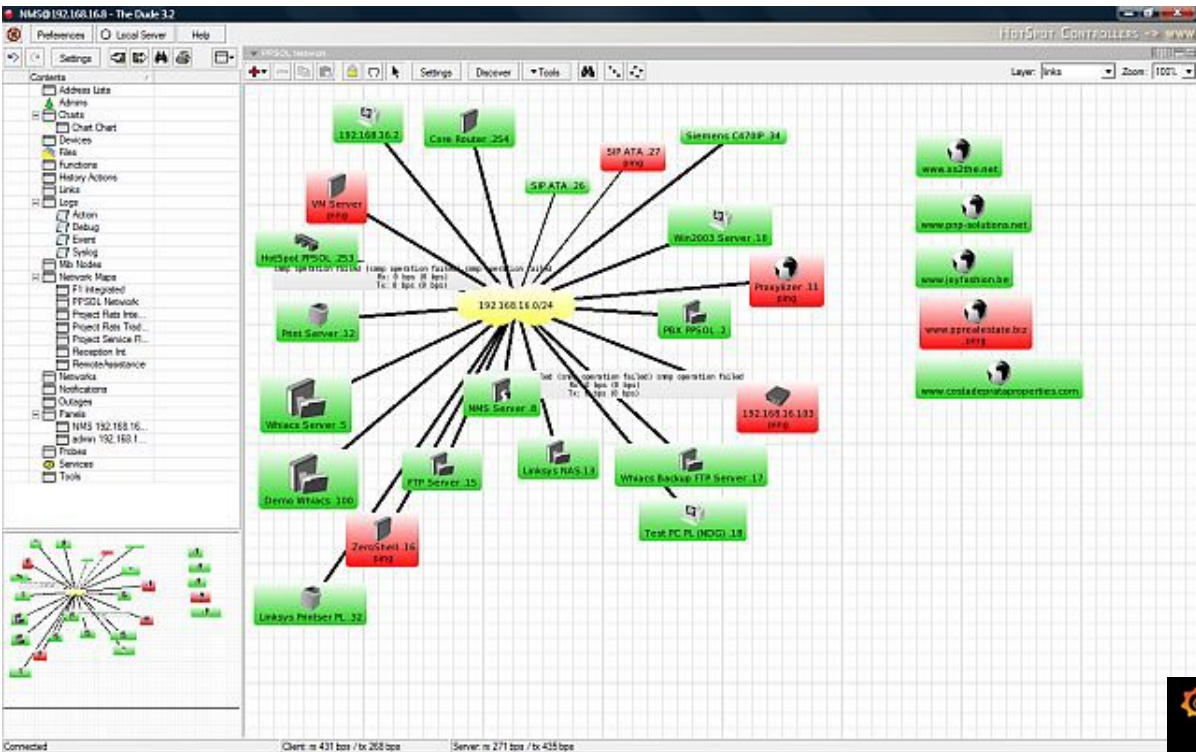


Maximizing Benefits with Azure: The Role of Monitoring

- One of the many reasons you may be hosting your infrastructure or applications on Azure, **may be performance.**
- The deployment of resources shouldn't be the end of your interaction.
 - How do you know if you are gaining benefits with Azure hosting?
 - How can you prove your return on investment?
 - Are there aspects to your applications that are not giving the performance you, or your users, expect?
- Monitoring your resources **provides insights into many aspects of performance, security, and health.**

Monitoring

-  Alerts
-  Metrics
-  Logs
-  Health check
-  Diagnostic settings (preview)
-  App Service logs
-  Log stream





Example scenario

You decided to move your web application **from your on-premises infrastructure** to Azure due to performance issues.

- During back-to-school campaigns, your web site is being overloaded and response times are causing users to leave.

You don't want to spend more money on new server hardware just to support these limited engagements where load peaks.

Azure offers the ability to scale up, or scale out, your applications, when needed. Azure can then automatically scale them back when resource needs drop off. Initially, you may not know where the capacity issues are.



On-premises
data center

Migration
Process



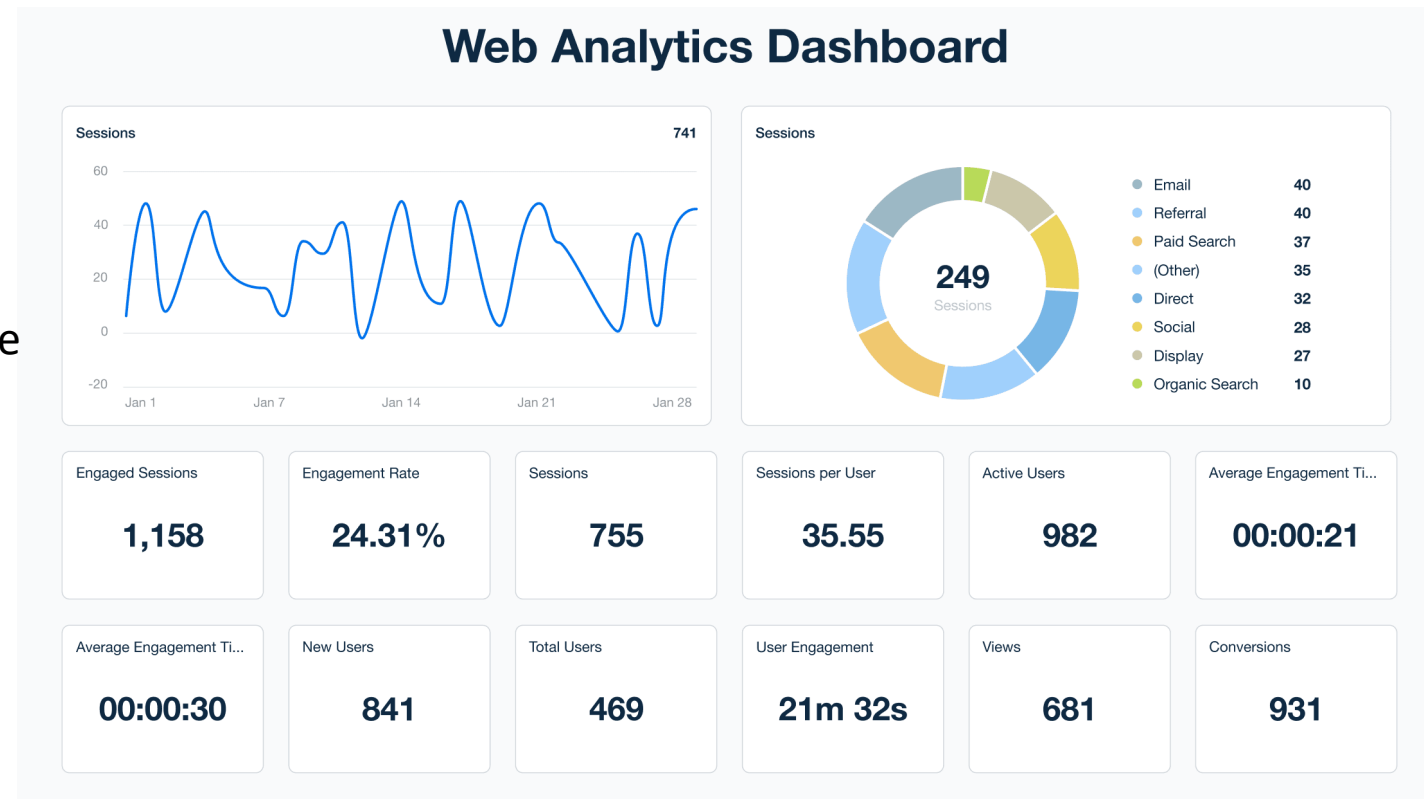
Cloud
architecture

Using the Azure Monitor capabilities, you can determine where your application scaling needs are and adjust those parameters in your scaling strategy. You can base these decisions on concrete metrics and log data.

How to monitor your web application

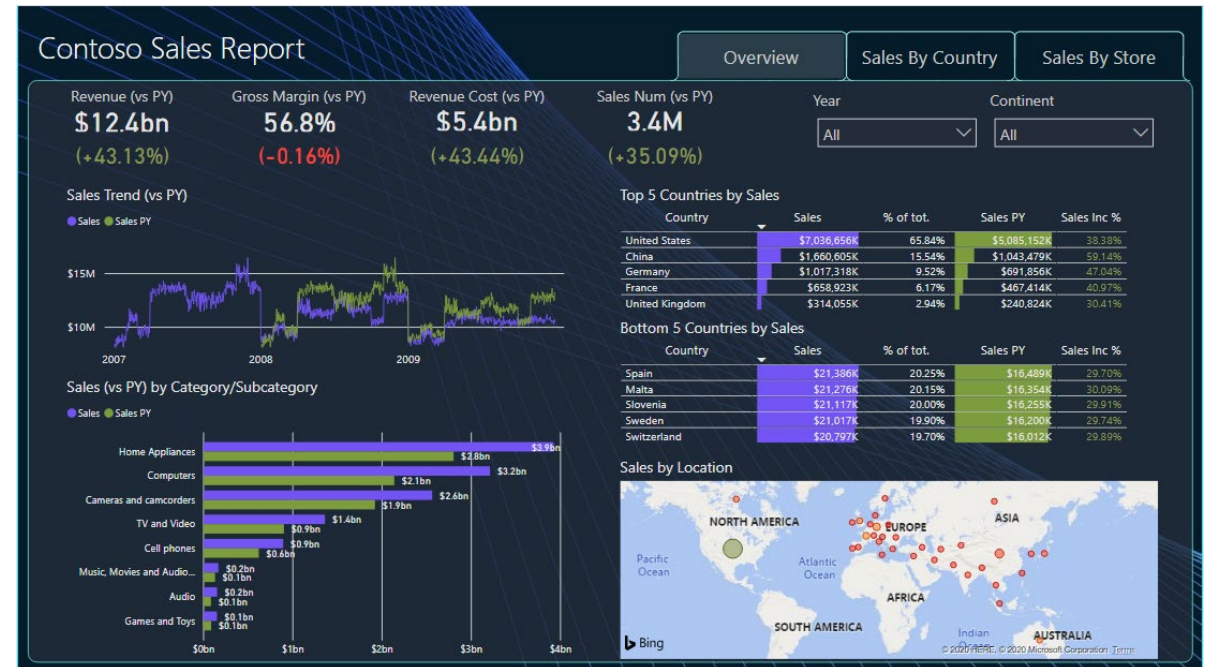
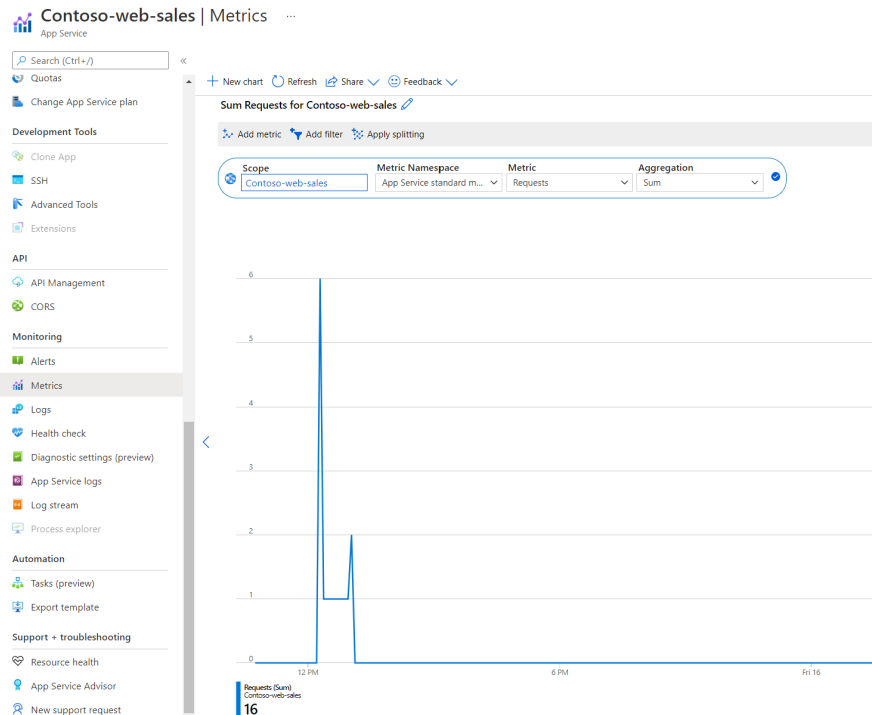
Evaluating the performance of a web application requires monitoring where, and when, the web app experiences demand.

Azure Monitor is enabled by default. As soon as you create an Azure subscription and start adding resources, such as web apps, Azure Monitor starts collecting data. Activity logs record when resources are created or modified. You can use the information in the activity logs to trace the history of the resource creation and modifications. Metrics tell you how the resource is performing and any other resources that it's consuming.



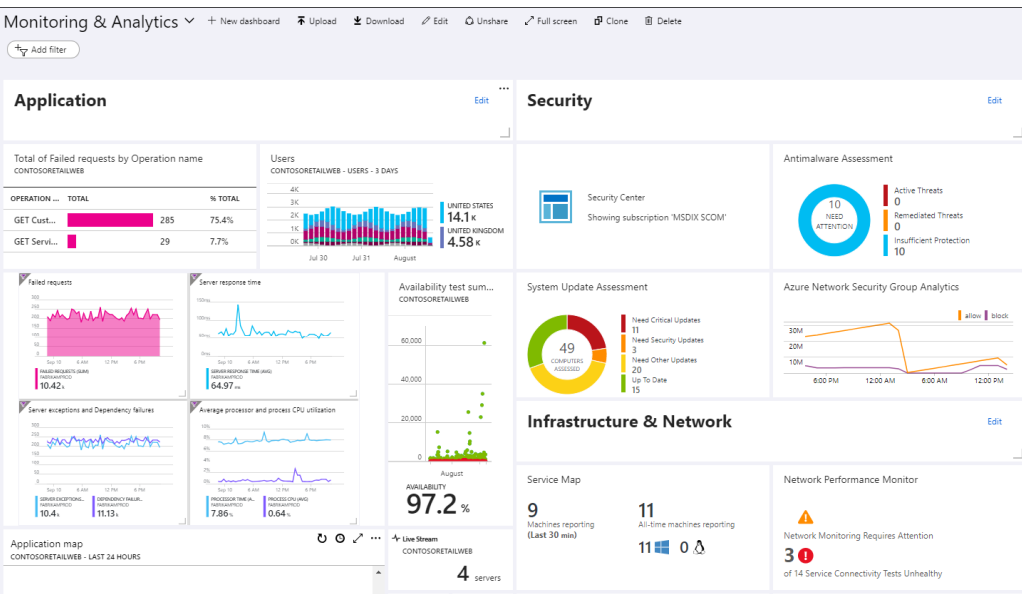
How to monitor your web application

- As an example, the **Contoso-web-sales Metrics** chart displayed below, demonstrates the ability to select the **Requests** metric for web app. The chart displays the sum aggregation of the requests for the web app, using a line chart format. The list of metrics is available automatically for your resource. There is no requirement to install other software or services. You find Metrics under the Monitoring category in each resource panel that you have in your Azure subscription.



Other monitoring considerations

- **Interactive monitoring** is just one way to utilize the features of Azure Monitor. Many situations can benefit from a strategy involving alerts. You can configure alerts that will proactively respond to critical conditions.
- These alerts can be sent over text message or email, to have a person investigate. You may also consider configurations that can take actions based on an alert. Using **action groups**, which contain a set of recipients and actions to take, you can simplify administration across multiple rules.



Other monitoring considerations

- **Workbooks** provide a flexible canvas for data analysis and the creation of rich visual reports in the Azure portal.
- They allow you to tap into multiple data sources from across Azure, and combine them into unified interactive experiences. Use workbooks provided with Insights or create your own from predefined templates.





Your channel
Hootsuite Labs



Dashboard



Content



Analytics



Comments



Subtitles



Copyright



Earn



Customization



Audio library



Settings

Channel analytics

Jan 1 – 26, 2023
January

Overview

Content

Audience

Revenue

Research

All

Videos

Shorts

Posts

Views

107.1K

11% more than Dec 6 – 31, 2022

Impressions

173.2K

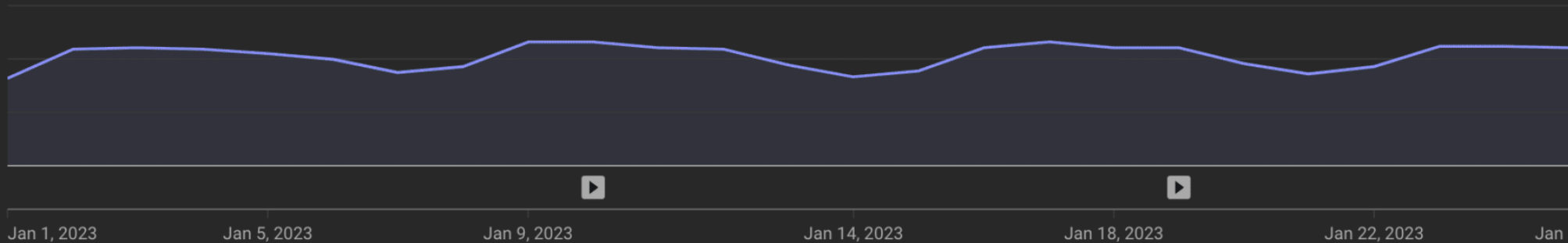
12% more than Dec 6 – 31, 2022

Impressions click-through rate

8.2%

Average view d

2:21



SEE MORE

Key moments for audience retention

Latest videos (last 365 days)

Intro

Top moments

Sp

Azure Monitoring and on-premises services

Azure Monitoring aggregates and **analyse metrics, logs, and traces**. Fire **alerts and send notifications** or call **automated solutions**.

It delivers a comprehensive solution for collecting, analysing, and acting on telemetry from your cloud and on-premises environments.

This information helps you understand how your applications are performing and proactively identify issues affecting them and the resources they depend on.

A few examples of what you can do with Azure Monitor include:

- Detect and diagnose issues across applications and dependencies with **Application Insights**.
- Correlate infrastructure issues with **VM insights** and **Container insights**.
- Drill into your monitoring data with **Log Analytics** for troubleshooting and deep diagnostics.
- Support operations at scale with **smart alerts** and **automated actions**.
- Create visualizations with **Azure dashboards** and **workbooks**.
- Collect data from **monitored resources** using **Azure Monitor Metrics**.



Azure Monitor





Azure Monitor

Definition

Telemetry: the process of recording and transmitting the readings of an instrument.



Azure Monitor is a service for **collecting, analysing, and acting on telemetry from your cloud and on-premises environments**. Azure Monitor is used by IT Operations, DevOps, and developer teams to maximize the availability and performance of applications and services.



Azure Monitor **provides a broad set of capabilities** with a varying set of prerequisites depending on your specific monitoring goal.



On one end of the spectrum, getting started with Azure Monitor metrics requires no incoming experience because **metric collection is on by default for Azure resources**.

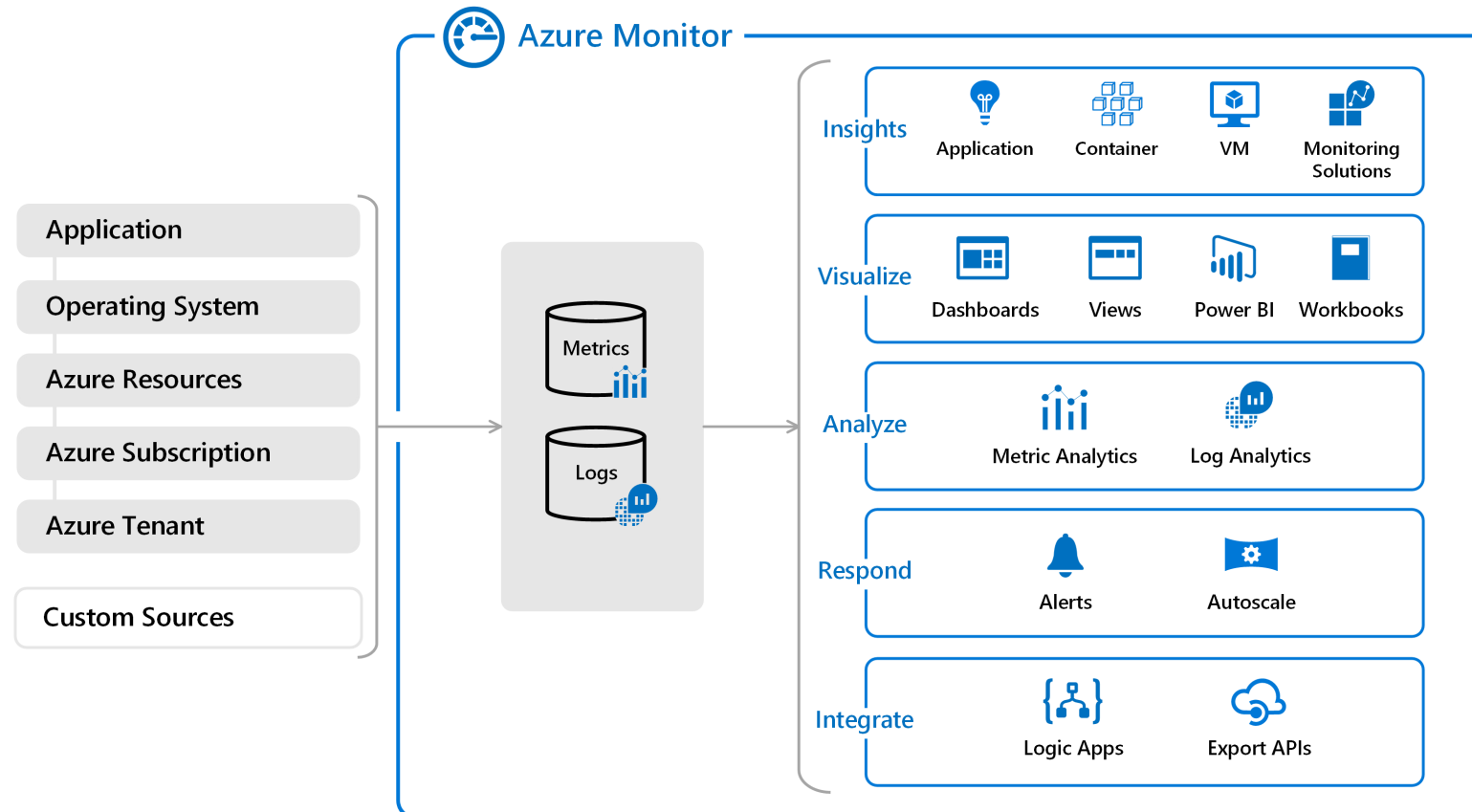


Viewing metrics in Metric Explorer is **straightforward in the Azure portal UI**. However, when using Azure Monitor Log Analytics for trend analysis and pattern recognition, **familiarity with query languages would be helpful**.

Azure Monitor: Key Aspects

Key aspects of Azure Monitor are:

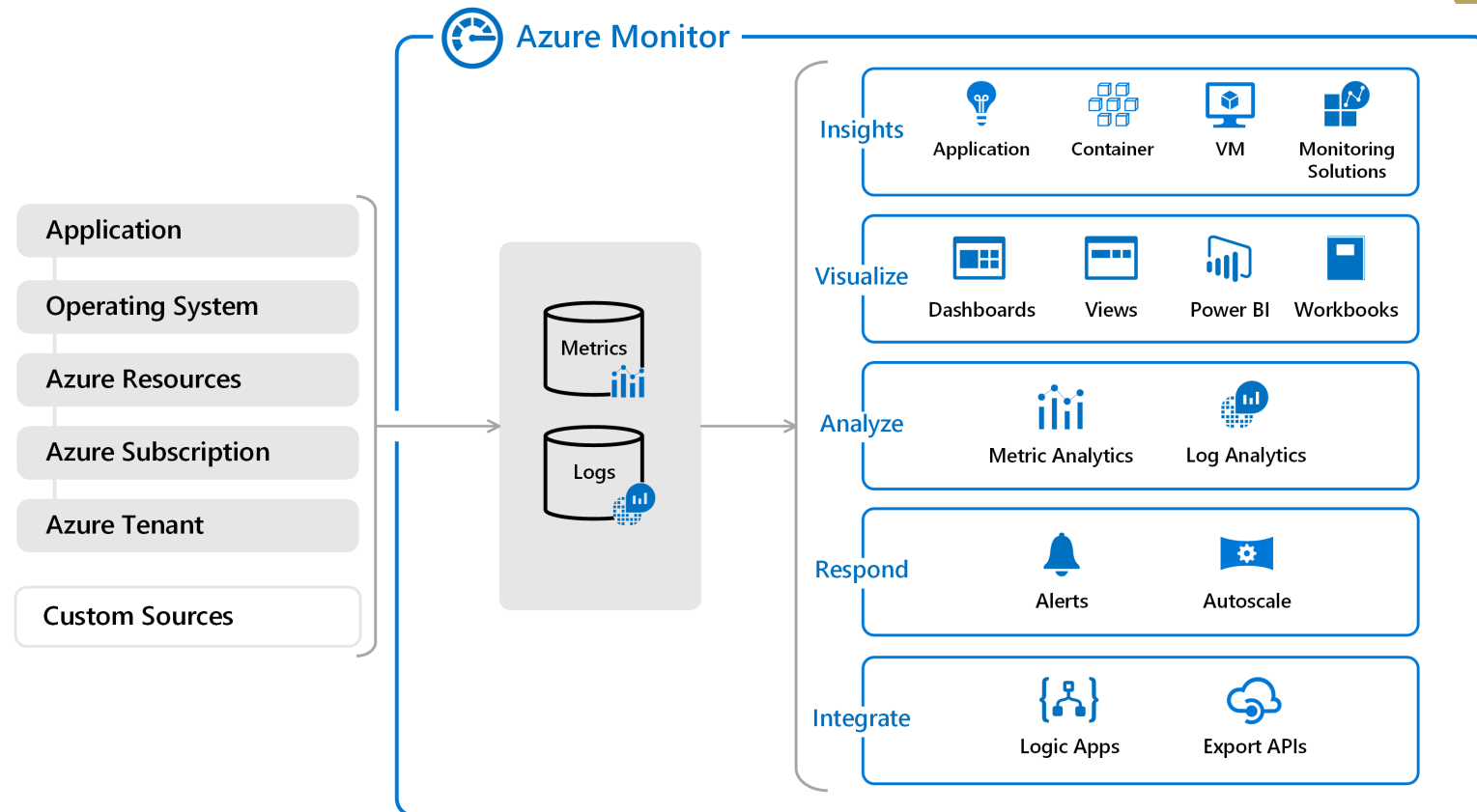
- Azure Monitor can **collect performance and availability telemetry** for all of the layers in your stack, **from applications, infrastructure, the Azure platform, and any custom sources you integrate.**



Azure Monitor: Data Collection and Storage

Key aspects of Azure Monitor are:

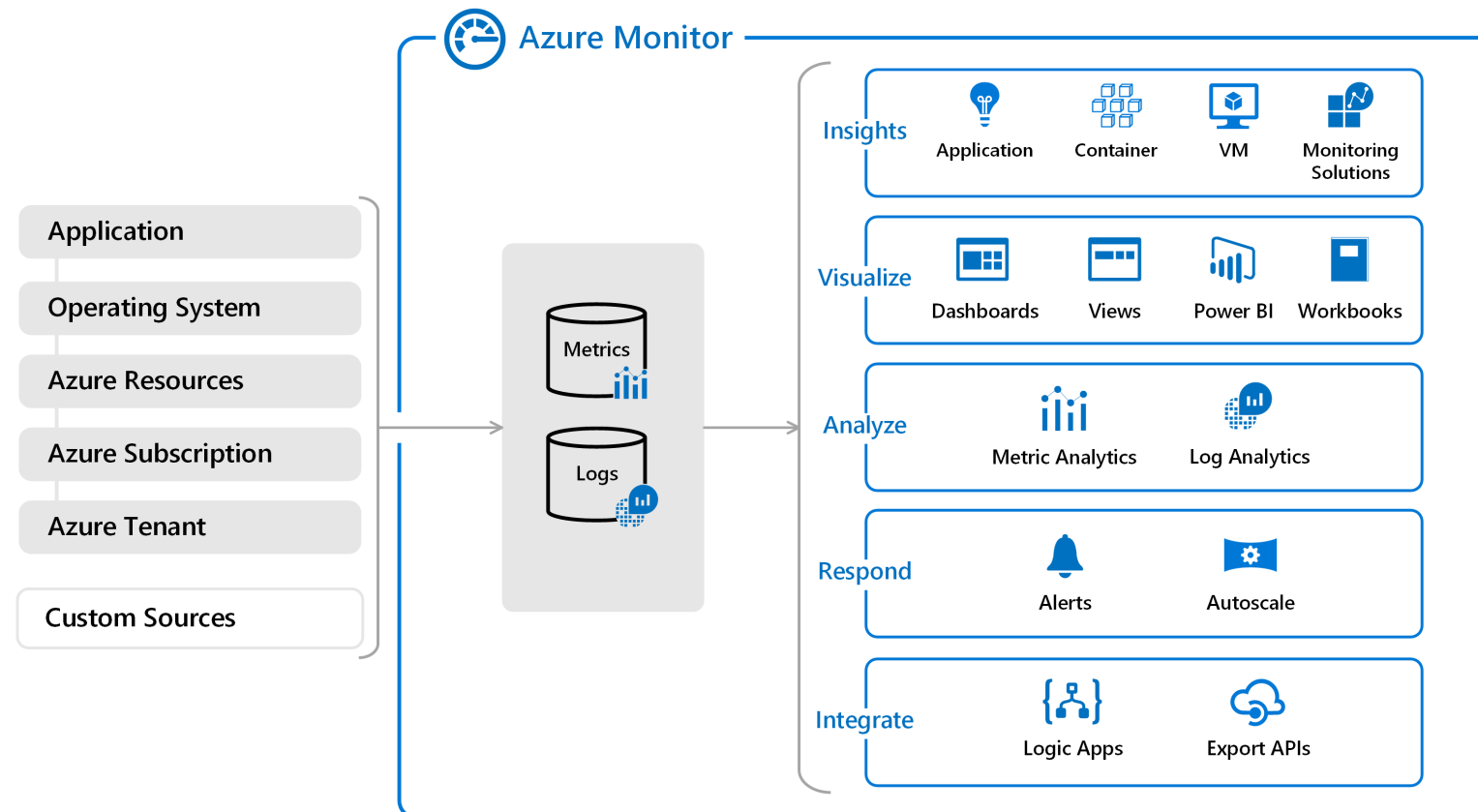
- Storage of the collected data in one of two data stores, **Azure Monitor Metrics** for numerical time-series values and **Azure Monitor Logs** for storing log data.



Azure Monitor: Metrics vs. Logs Explained

Key aspects of Azure Monitor are:

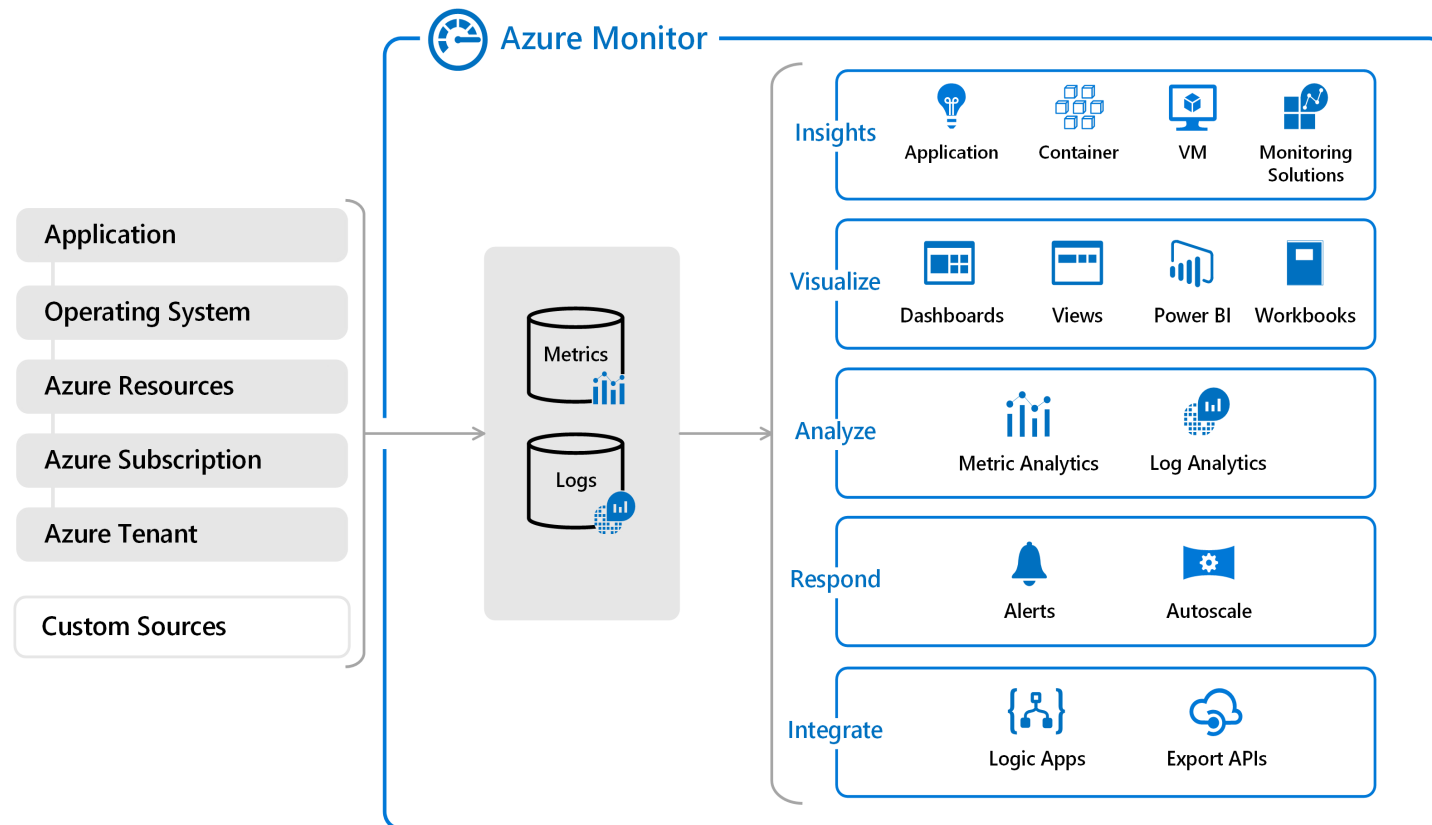
- Azure Monitor **metrics** are automatically collected and stored for Azure **resources**, but **user configuration is required** to send and **store resource logs** in Azure Monitor Logs.



Azure Monitor Insights: Built-In Solutions for Resource Monitoring

Key aspects of Azure Monitor are:

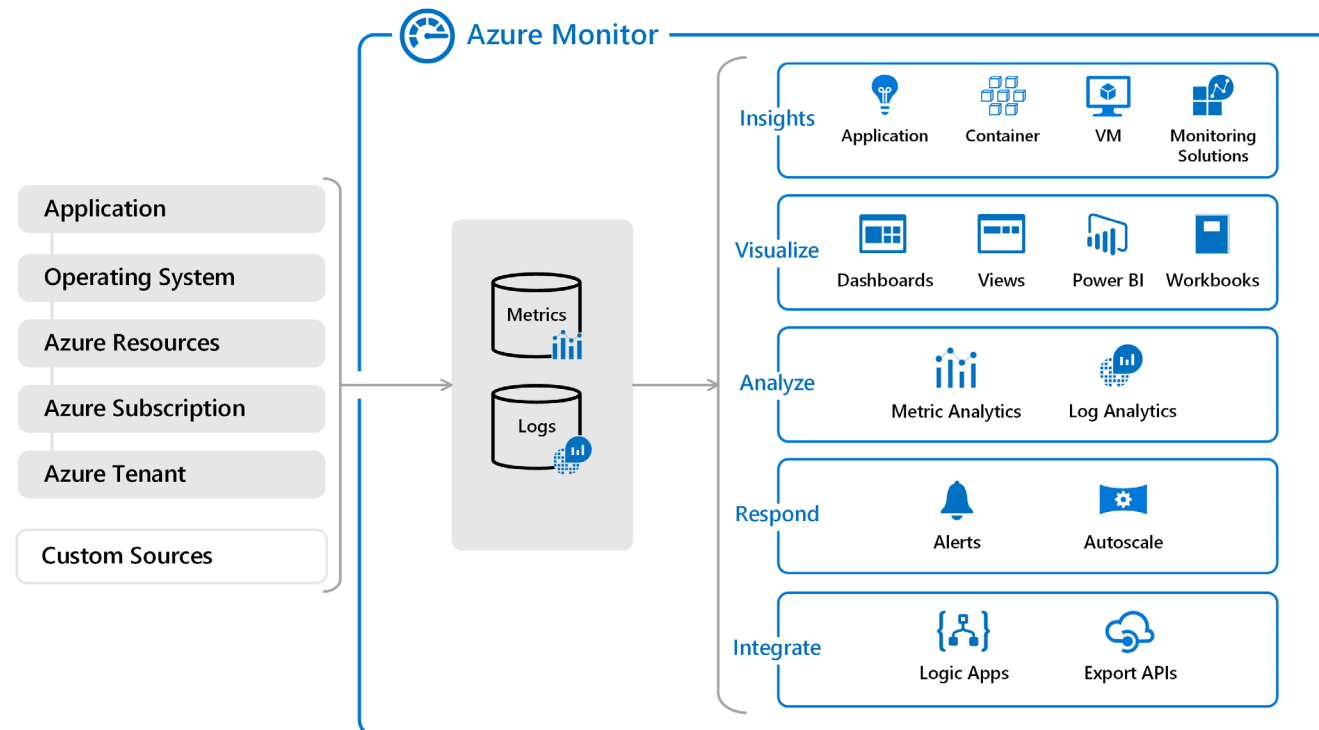
- Azure Monitor includes **Insights**, which are **out-of-the box monitoring and troubleshooting experiences for Azure resources**. Available Insights include Azure Monitor VM insights, Azure Monitor application insights, and Azure Monitor container insights.



Azure Monitoring: Visualise, Analyse, and Automate

Key aspects of Azure Monitor are:

- Azure Monitor can be used to **visualize data with workbooks and dashboards as well as analyse data with custom charts and analytics.**
- Azure Monitor enables you to **receive notification and take automated action based on performance** and availability criteria.



Getting Started with Azure Monitor: Tools and Features

You can monitor and start investigations in Azure Monitor directly in the Azure portal or access Azure Monitor capabilities in the Monitoring section of your Azure resource blade.

You can enable additional Azure Monitor features for your resources at-scale programmatically with Azure CLI, Azure Resource Manager (ARM) templates, and PowerShell. You can ingest and export data via APIs and you can use Azure Event Hubs to stream data out in near real time.

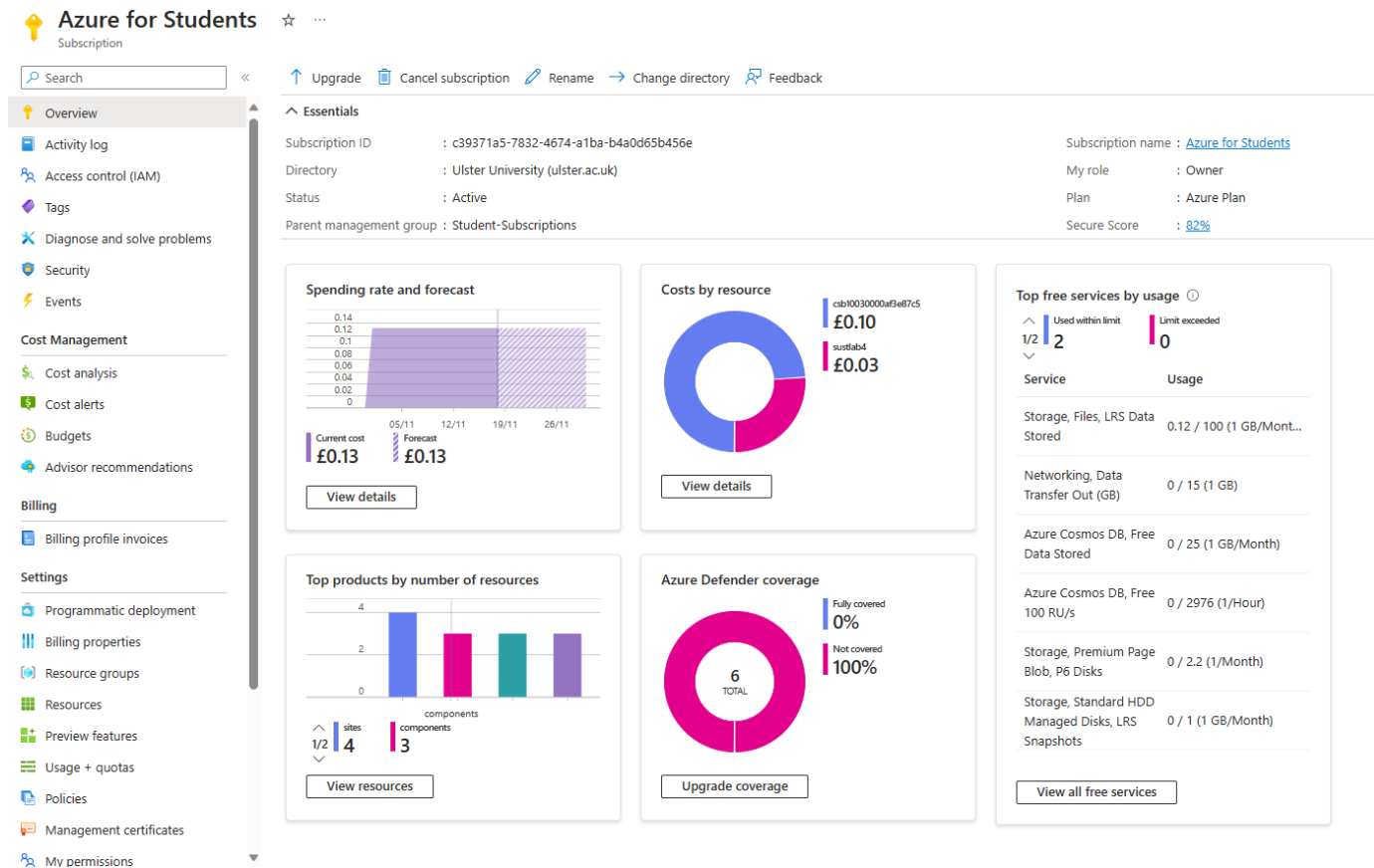
You'll learn how Azure Monitor can:

- Display metrics for your services
- How Azure Monitor agents are used with compute resources
- How Azure Monitor makes use of logs for data storage

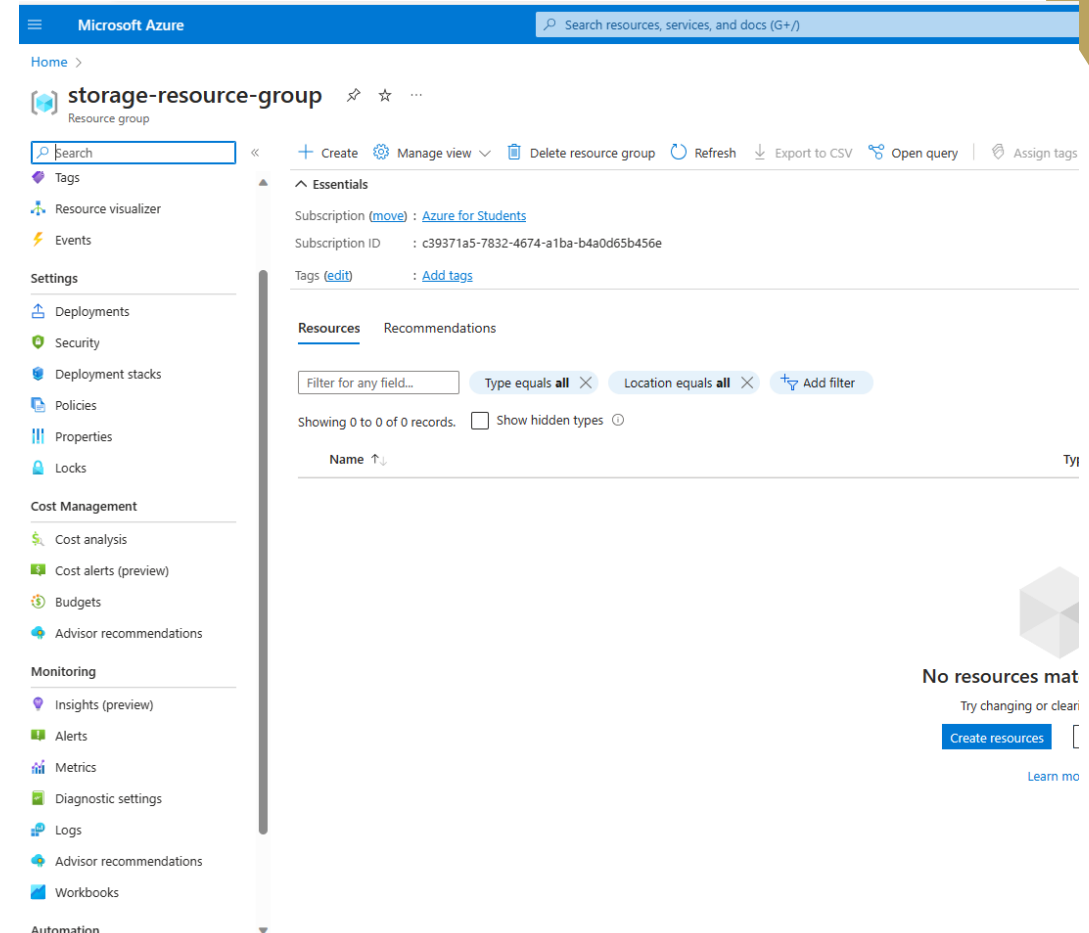
* We will do
practical using
Azure monitoring
in labs

Azure Monitor metrics

You can find **Metrics** under the **Monitoring** category of the resource pane in the Azure portal. Once you select **Metrics**, you will be able to select the **Metric Namespace**, **Metric** category, and **Aggregation** from the drop-down menu choices above the graph.

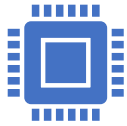


The screenshot shows the 'Azure for Students' subscription page. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Security, Events, Cost Management (Cost analysis, Cost alerts, Budgets, Advisor recommendations), Billing (Billing profile invoices), Settings (Programmatic deployment, Billing properties, Resource groups, Resources, Preview features, Usage + quotas, Policies, Management certificates, My permissions), and Upgrade. The main content area is titled 'Essentials' and displays subscription details: Subscription ID (c39371a5-7832-4674-a1ba-b4a0d65b456e), Directory (Ulster University (ulster.ac.uk)), Status (Active), Parent management group (Student-Subscriptions), Subscription name (Azure for Students), My role (Owner), Plan (Azure Plan), and Secure Score (82%). Below this are four charts: 'Spending rate and forecast' (a bar chart showing current cost of £0.13 and a forecast of £0.13), 'Costs by resource' (a donut chart showing £0.10 for 'c3b10030000af9e87c5' and £0.03 for 'sustlab4'), 'Top free services by usage' (a table listing services like Storage, Files, LRS Data Stored, Networking, Data Transfer Out (GB), Azure Cosmos DB, Free Data Stored, Azure Cosmos DB, Free 100 RU/s, Storage, Premium Page Blob, P6 Disks, Storage, Standard HDD Managed Disks, LRS Snapshots, and their usage), and 'Azure Defender coverage' (a donut chart showing 0% fully covered and 100% not covered for 6 total components).



The screenshot shows the 'storage-resource-group' resource page. The left sidebar contains navigation links: Home, storage-resource-group, Tags, Resource visualizer, Events, Settings (Deployments, Security, Deployment stacks, Policies, Properties, Locks), Cost Management (Cost analysis, Cost alerts (preview), Budgets, Advisor recommendations), Monitoring (Insights (preview), Alerts, Metrics, Diagnostic settings, Logs, Advisor recommendations, Workbooks), and Automation. The main content area is titled 'Essentials' and displays resource details: Subscription (move) (Azure for Students), Subscription ID (c39371a5-7832-4674-a1ba-b4a0d65b456e), Tags (edit), and Add tags. Below this is a 'Resources' section with a filter bar (Filter for any field..., Type equals all, Location equals all, Add filter) and a message 'Showing 0 to 0 of 0 records. Show hidden types'. At the bottom right, there is a 'No resources mat' message with a 'Create resources' button and a 'Learn mo' link.

Azure Monitor agents



Virtual machines and other compute resources require an agent to collect monitoring data required to measure the performance and availability of their guest operating system and workloads.



Centrally configure data collection for different data from different sets of virtual machines.



The Azure Monitor agent (AMA) collects monitoring data from the guest operating system of virtual machines and delivers it to Azure Monitor.



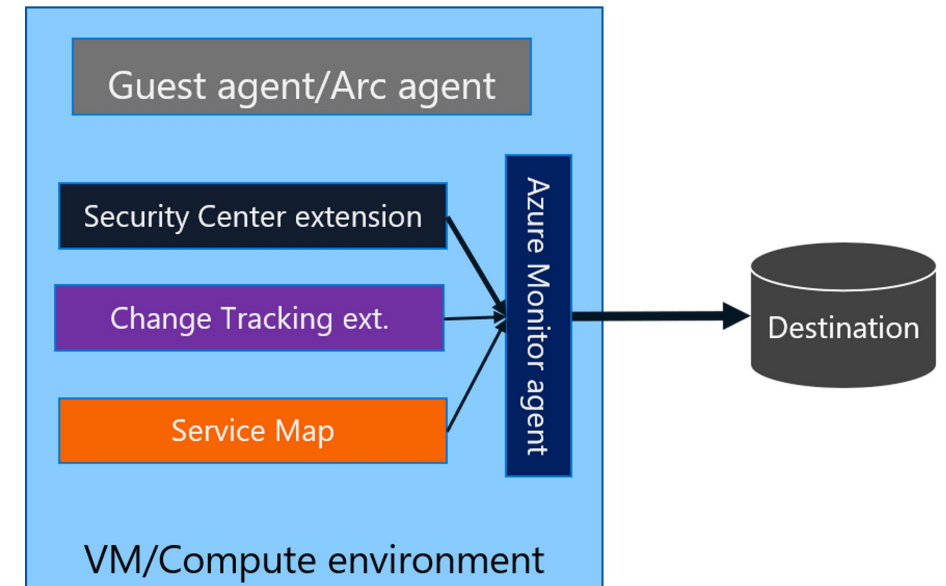
Send data from Linux virtual machines to multiple workspaces.



The Azure Monitor agent has replaced the Log Analytics agent, the diagnostic extension, and the Telegraf agent. By utilizing the Azure Monitor Agent, you can realize the following benefits:

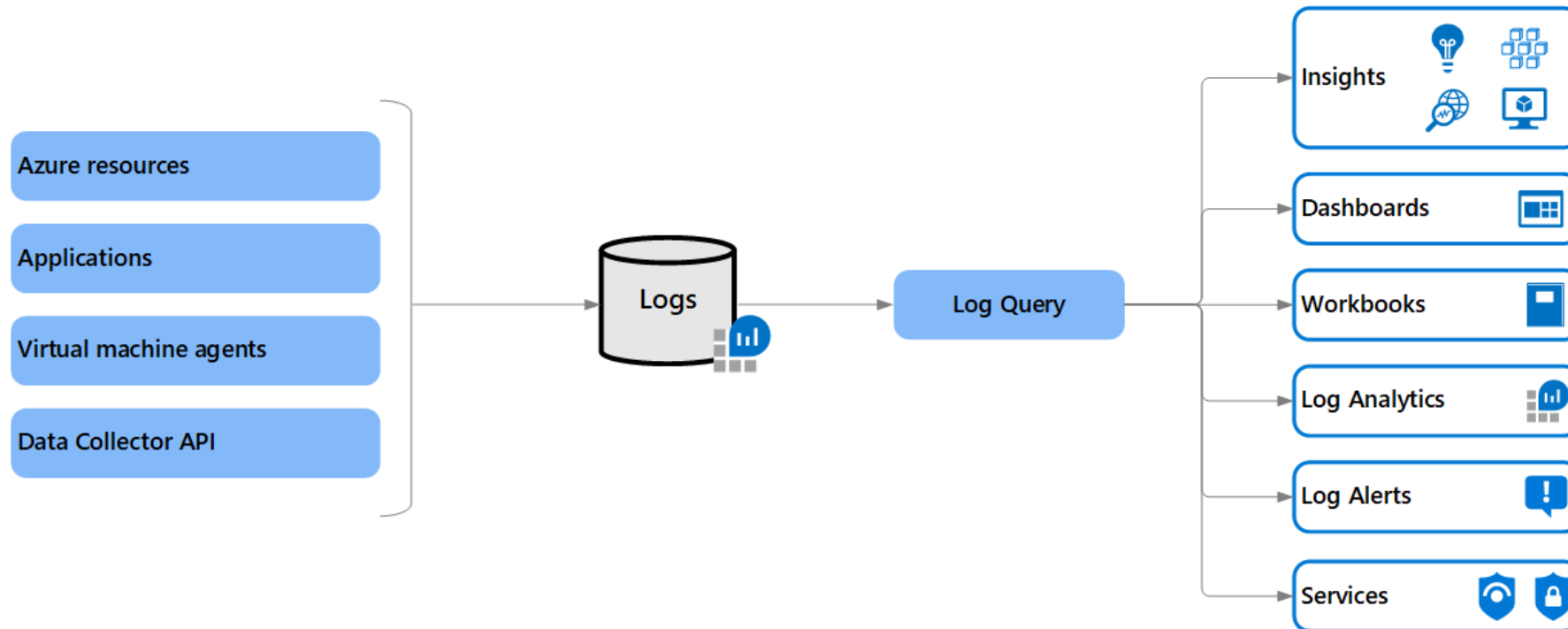


Use XPATH queries to filter Windows events for collection.



Azure Monitor log: Data Flow Overview

In the log overview image below, data sources on the left indicate Azure resources, applications, virtual machine agents, and the data collector API, feeding into the log database in the middle. Using a log query, you can output to the sources on the right which indicate Insights, Dashboards, workbooks, log analytics, alerts, and other services.



Azure Monitoring: Metrics vs. Logs for Data Analysis

Azure Monitor Metrics allows you to view **time-series data in near real-time and retains data for up to three months.**

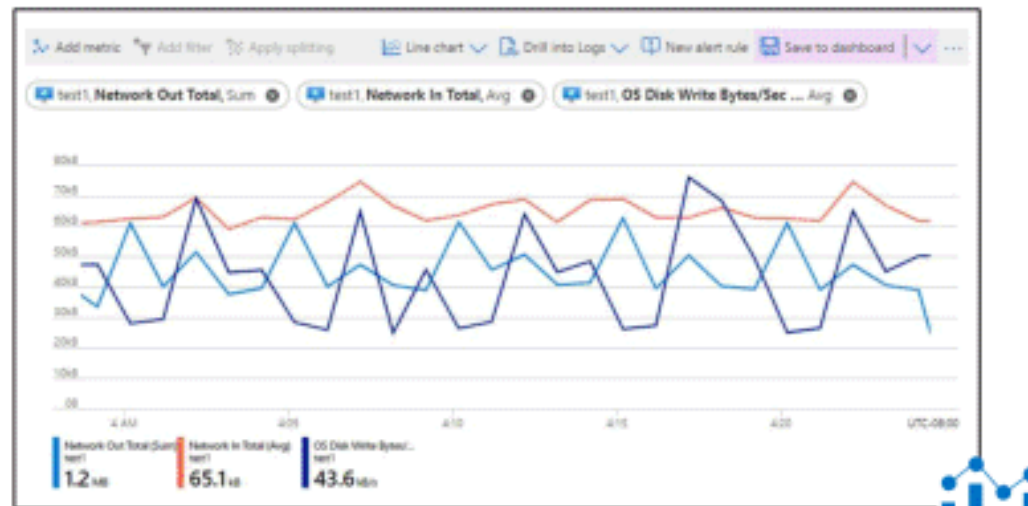
If you want to **store data for long-term evaluation up to two years and perform more complex analysis**, you would use a Log Analytics Workspace to store the recorded data.

Let's look at some of the different mechanisms for monitoring and viewing your Azure resources.

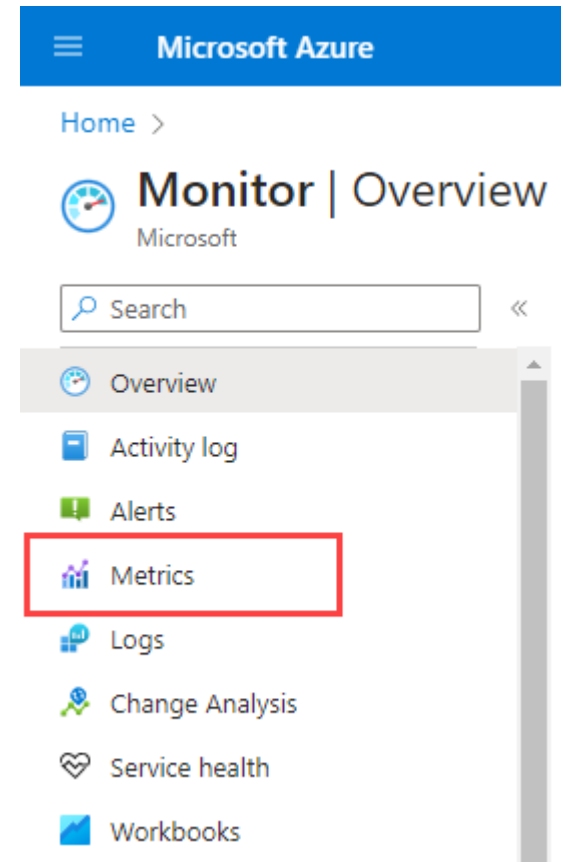
Monitoring Option	Usage
Metrics	Numerical values that describe some aspect of a system at a particular point in time.
Logs	Log data collected by Azure Monitor for analysis.
Visualizations	Azure Monitor has its own features for visualizing monitoring data and uses other Azure services for publishing it to different audiences. Azure dashboards allow you to combine different kinds of data into a single pane in the Azure portal. Workbooks provide a flexible canvas for data analysis and the creation of rich visual reports in the Azure portal.

Metric

- Metrics are **numerical values that describe some aspect of a system at a particular point in time.**
- They are lightweight and capable of supporting near real-time scenarios.
- The metrics explorer is built into the Azure portal, allowing you to focus on a specific resource to monitor.
- Data is presented in a chart or graph format.



Metrics Explorer



Select a scope



Browse Recent

Subscription

AI_Demo_vitalyg_RnD

Resource types

All resource types

Locations

All locations

Search to filter items...

Filter items...

Scope

☐ AI_Demo_vitalyg_RnD

☐ > Default-Applicat

☐ > DefaultResourc

☐ > MetricsBaseline

☐ > NetworkWatch

☐ > Site-recovery-v

☐ > TailspinToys

☐ > TailspinToysDer

☐ > TailspinToysLin

☐ > Telemetry

☐ > cleanupservice

☐ > cloud-shell-sto

☒ Select all

Multi-resource compatible (preview)

☒ Azure Cache for Redis

☒ Azure Stack Edge / Data Box Gateway

☒ Capacity pools

☒ Key vaults

☒ SQL databases

☒ Virtual machines

All resource types

☒ Analysis Services

☒ Apache Spark pools

☒ API Management services

☒ App Service (Slots)

☒ App Service Environments

Location

-

-

-

-

-

-

-

-

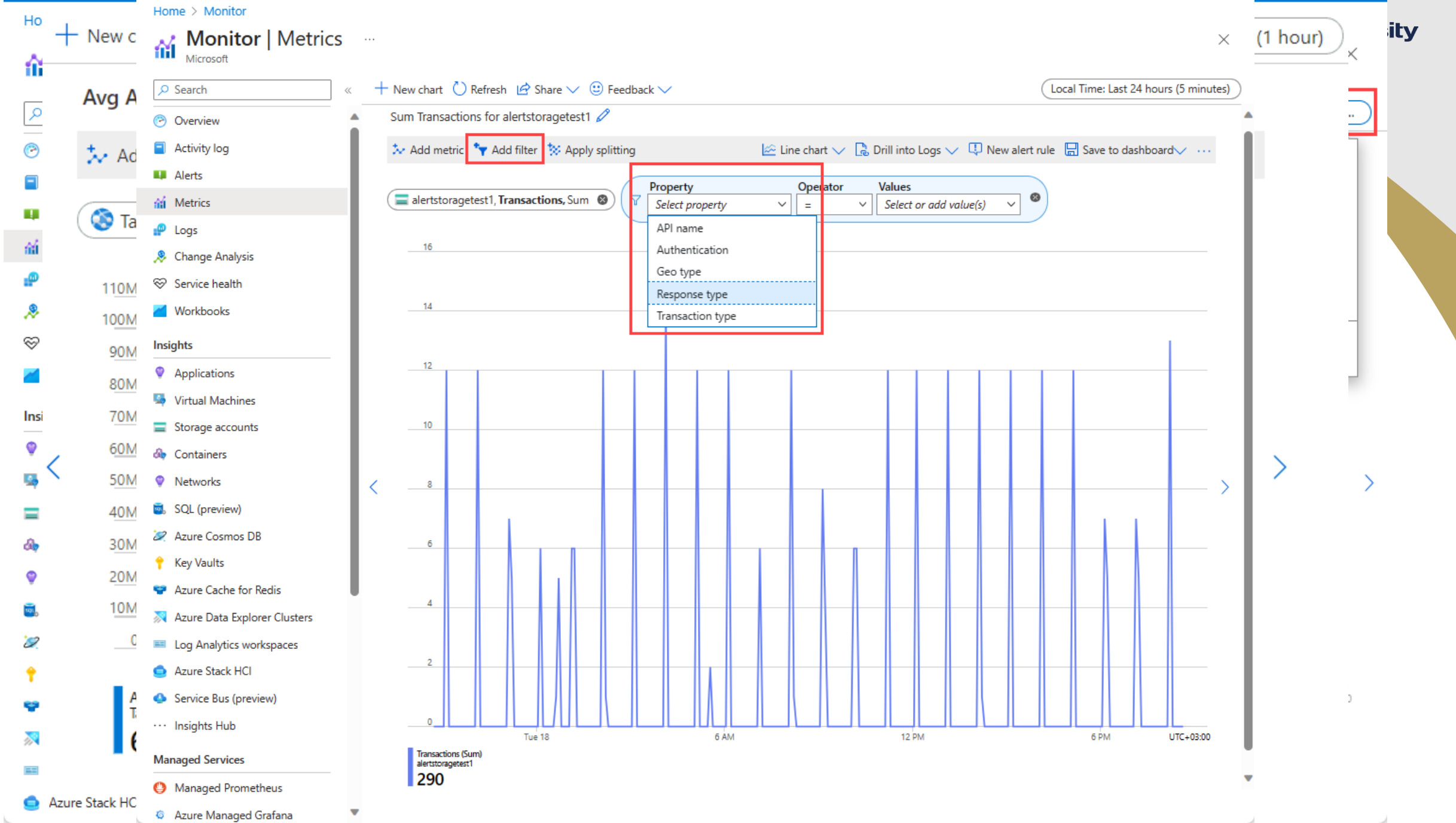
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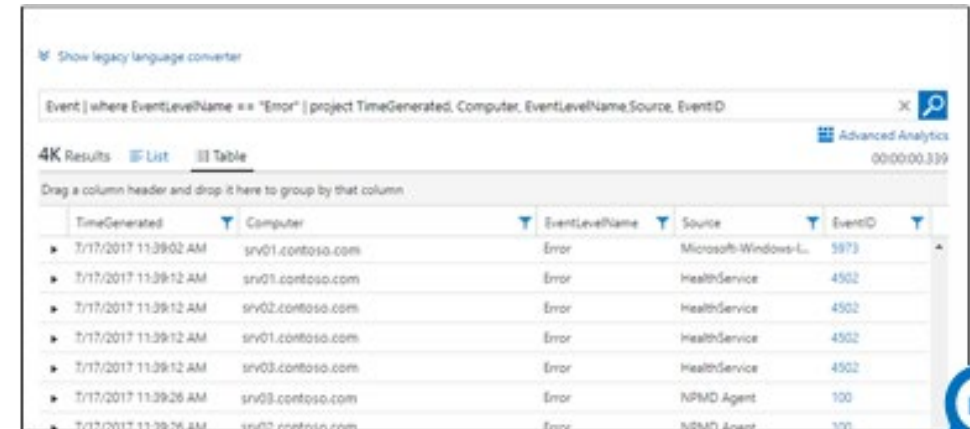
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Analyzing Logs with Azure Monitor: Querying and Insights

- Log data collected by Azure Monitor can be analysed with queries to quickly retrieve, consolidate, and analyse collected data.
- **You can create and test queries using Log Analytics in the Azure portal.** You can then either directly analyse the data using different tools or save queries for use with visualizations or alert rules.
- Azure Monitor uses a version of the **Kusto query language** that is suitable for **simple log queries** but also includes advanced functionality such as aggregations, joins, and smart analytics.



Event | where EventLevelName == "Error" | project TimeGenerated, Computer, EventLevelName, Source, EventID

4K Results

TimeGenerated	Computer	EventLevelName	Source	EventID
7/17/2017 11:39:02 AM	srv01.contoso.com	Error	Microsoft-Windows-L...	5873
7/17/2017 11:39:12 AM	srv01.contoso.com	Error	HealthService	4502
7/17/2017 11:39:12 AM	srv02.contoso.com	Error	HealthService	4502
7/17/2017 11:39:12 AM	srv01.contoso.com	Error	HealthService	4502
7/17/2017 11:39:12 AM	srv03.contoso.com	Error	HealthService	4502
7/17/2017 11:39:26 AM	srv03.contoso.com	Error	NPMD Agent	100
7/17/2017 11:39:56 AM	srv01.contoso.com	Error	NPMD Agent	100

Comparison of Metrics and Logs in Azure Monitor

The following table compares Metrics and Logs in Azure Monitor.

Attribute	Metrics	Logs
Benefits	Lightweight and capable of near-real time scenarios such as alerting. Ideal for fast detection of issues.	Analyzed with rich query language. Ideal for deep analysis and identifying root cause.
Data	Numerical values only	Text or numeric data
Structure	Standard set of properties including sample time, resource being monitored, a numeric value. Some metrics include multiple dimensions for further definition.	Unique set of properties depending on the log type.
Collection	Collected at regular intervals.	May be collected sporadically as events trigger a record to be created.
View in Azure portal	Metrics Explorer	Log Analytics
Data sources include	Platform metrics collected from Azure resources. Applications monitored by Application Insights. Custom defined by application or API.	Application and resource logs. Monitoring solutions. Agents and VM extensions. Application requests and exceptions. Azure Security Center. Data Collector API.

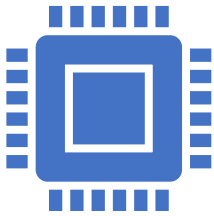
F1 Insights : Car Performance Scores



Azure Insights



Insights



Monitoring data is only useful if it can increase your visibility into the operation of your computing environment.



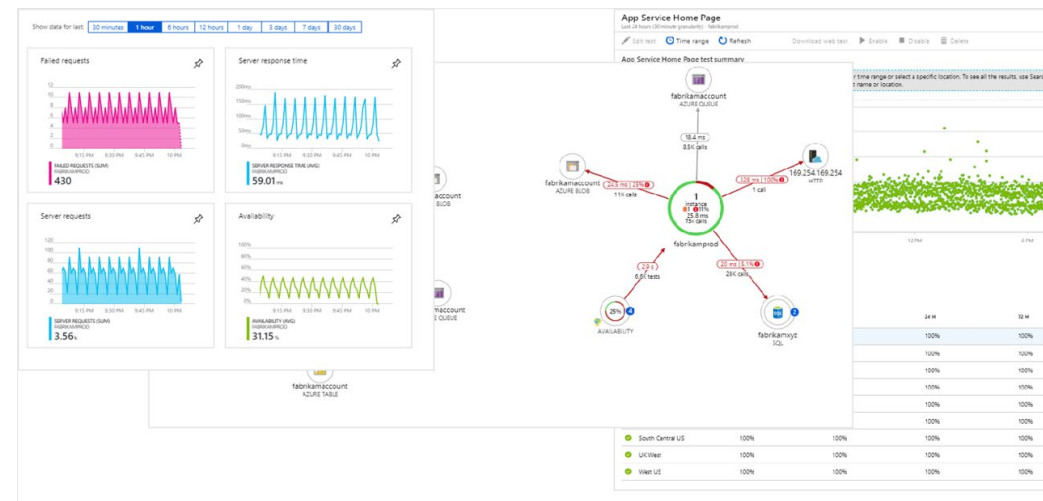
Insights provide a customized monitoring experience for particular Azure services.



They require minimal configuration and increase your visibility into the operation of critical resources.

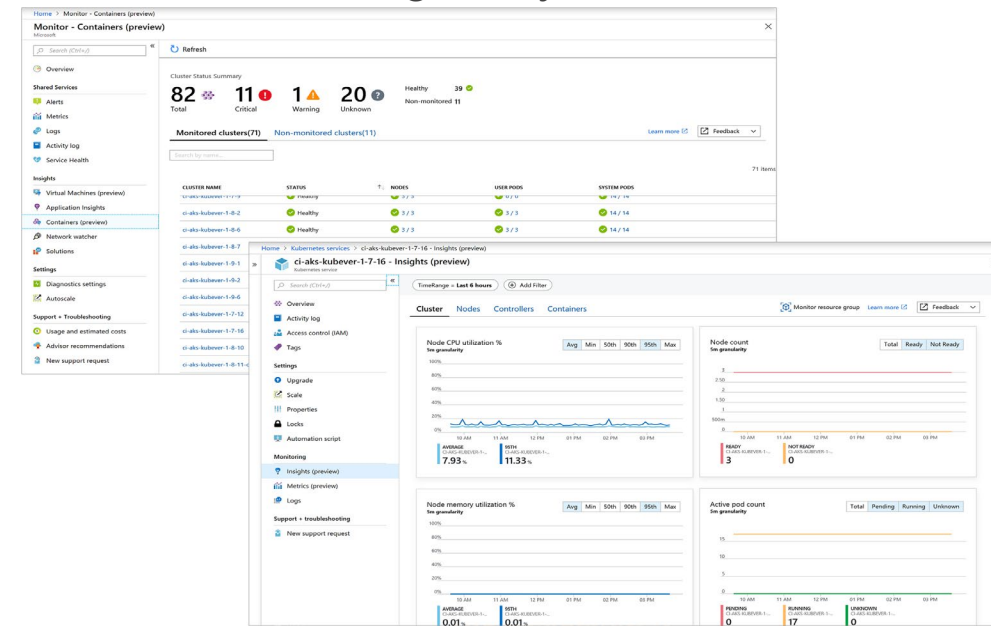
Application Insights monitors the availability, performance, and usage of your web applications whether they're hosted in the cloud or on-premises.

It enables you to diagnose errors without waiting for a user to report them. Application Insights includes connection points to a variety of development tools and integrates with Visual Studio to support your DevOps processes.

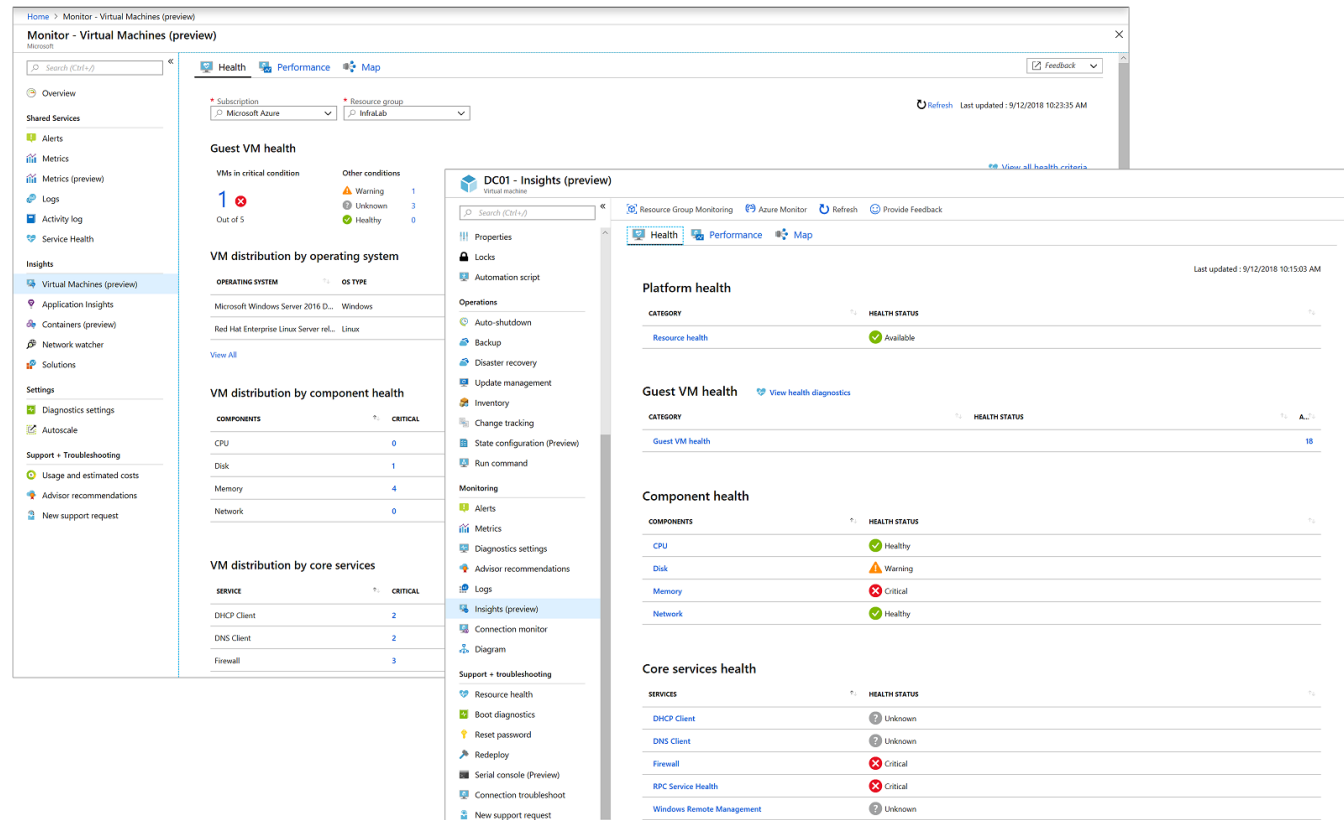


B) Container insights

- Container insights monitors the performance of container workloads that are deployed to managed Kubernetes clusters hosted on Azure Kubernetes Service (AKS).
- It gives you performance visibility by collecting metrics from controllers, nodes, and containers that are available in Kubernetes through the Metrics API. Container logs are also collected.
- After you enable monitoring from Kubernetes clusters, these metrics and logs are automatically collected for you through a containerized version of the Log Analytics agent for Linux.



- VM insights monitor your Azure virtual machines (VM) at scale. It analyses the performance and health of your Windows and Linux VMs and identifies their different processes and interconnected dependencies on external processes.
- The solution includes support for monitoring performance and application dependencies for VMs hosted on-premises or another cloud provider.



The screenshot displays the Azure Monitor VM Insights interface, which provides a comprehensive view of a virtual machine's health and performance. The interface is divided into several sections:

- Guest VM health:** Shows the overall health of the guest operating system. It includes a summary of VMs in critical condition (1), warning (1), unknown (3), and healthy (0) states. Below this, it lists VMs in critical condition, such as "Out of 5".
- VM distribution by operating system:** A table showing the distribution of VMs by operating system type.
- VM distribution by component health:** A table showing the distribution of VMs by component health, including CPU, Disk, Memory, and Network.
- VM distribution by core services:** A table showing the distribution of VMs by core services, including DHCP Client, DNS Client, Firewall, and Windows Remote Management.
- Platform health:** A section showing the health of the platform, including Resource health, Guest VM health, and Component health.
- Core services health:** A section showing the health of core services, including DHCP Client, DNS Client, Firewall, RPC Service Health, and Windows Remote Management.

The interface also includes a sidebar with navigation options such as Overview, Alerts, Metrics, Logs, Activity log, Service Health, Insights, Virtual Machines (preview), Application Insights, Containers (preview), Network watcher, Solutions, Settings, Diagnostics settings, Autoscale, Support + Troubleshooting, Usage and estimated costs, Advisor recommendations, and New support request.

Responding to critical situations

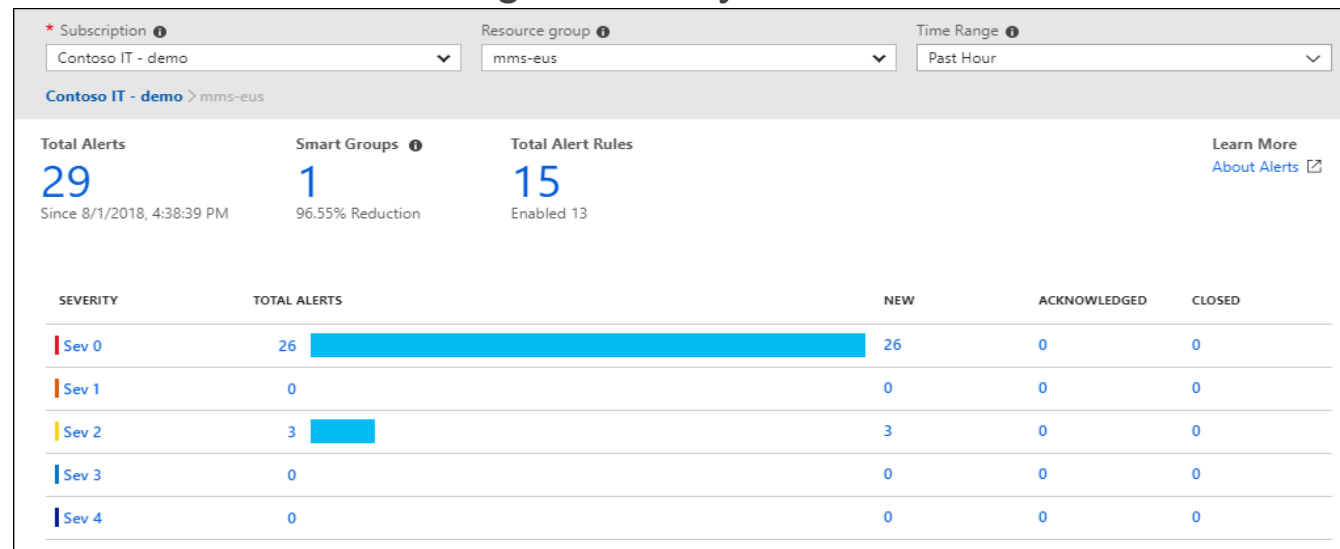
- An effective monitoring solution must be able to proactively respond to critical conditions identified in the data that it collects.
- This could be sending a text or mail to an administrator responsible for investigating an issue. Or you could launch an automated process that attempts to correct an error condition.



Responding to critical situations

Alerts

- Alerts in Azure Monitor proactively notify you of critical conditions and potentially attempt to take corrective action. Alert rules based on metrics provide near real time alerts based on numeric values. Rules based on logs allow for complex logic across data from multiple sources.
- Alert rules in Azure Monitor use [action groups](#), which contain unique sets of recipients and actions that can be shared across multiple rules. Based on your requirements, action groups can perform such actions as using webhooks to have alerts start external actions or to integrate with your ITSM tools.



Home > Cost Management: Zeeshan Tariq

Cost Management: Zeeshan Tariq | Cost analysis

Billing account



What are my forecasted costs for this month?

What were my total costs last month?

Can you show how my cost changed from

Search



Scope: Zeeshan Tariq (change)

- Overview
- Change scope
- Access control
- Diagnose and solve problems

> Reporting + analytics

Monitoring

Cost alerts

Alert rules

Budgets

> Optimization

> Settings

> Billing

> Help

New tab

Go to billing account Refresh Try preview Feedback ...

Analyze cloud costs and gain insights to understand and control spending. Views break down costs by resources, subscriptions, products, and services to help understand what you're spending

Recent All views Settings

Search

Recent [Browse all views](#)



Daily costs

Recommended



Accumulated costs



Resource groups



Subscriptions



Visualize your costs
[Analyze with charts](#)



Looking for more?
[Browse all views](#)

Prefer the previous experience? [Go to the Accumulated costs view](#)

\$

Cost Management: Zeeshan Tariq | Cost alerts

Billing account

Search

+

Add

Alert rules

Refresh

Dismiss

Re-Activate

Feedback

Help

- Overview
- Change scope
- Access control
- Diagnose and solve problems
- > Reporting + analytics
- > Monitoring
- \$

Cost alerts
- Alert rules
- Budgets
- > Optimization
- > Settings
- > Billing
- > Help

Scope : Zeeshan Tariq

\$

No Alerts to display

Your alerts will be shown here

Create alert rule

Zeeshan Tariq (Billing account)

Subscribe to alerts and get notified about changes. [Learn more](#)

Condition

Alert type *

Select an alert type

Create budget

Budget

* Alert conditions

Type	% of budget	Amount	
Actual cost	100	20	
Select type	Enter %	-	

* Alert recipients (email)

Alert recipients (email)

z.tariq@ulster.ac.uk





example@email.com

It is recommended to add azure-noreply@microsoft.com to your email allow list to ensure alert mails do not go to your spam folder.

Language preference

Select your preferred language for receiving the alert email for all recipients provided above. Default is the language associated to your enrollment.

Languages *

Default

Home > Cost Management: Zeeshan Tariq | Cost alerts >

 **Budgets**  ... 

What are my top 5 services with highest cost

What are some cost optimization strategies?

What's my current month's cost by service type?



 Add  Refresh  Feedback  Help 

Scope :  **Zeeshan Tariq**

 Search by name

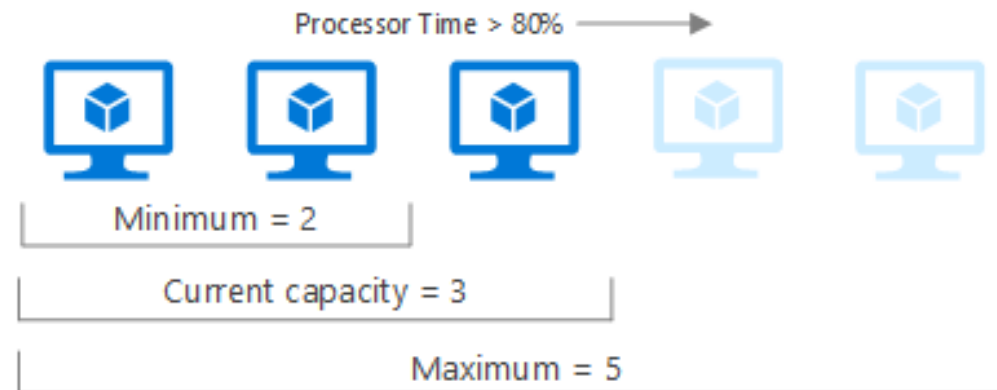
All periods 

Name	↑↓	Scope	↑↓	Reset period	↑↓	Creation date	↑↓	Expiration date	↑↓	Budget	↑↓	Forecasted	↑↓	Evaluated spend	↑↓	Progress	↑↓
COM682Budget		7c9ebb75-840c-425...		Monthly		11/1/2025		6/30/2026		\$20.00				\$0.00		0.00%	

Responding to critical situations

Auto-scale

- Auto-scale allows you to have the right amount of resources running to handle the load on your application. Create rules that use metrics collected by Azure Monitor to determine when to automatically add resources when load increases. Save money by removing resources that are sitting idle. You specify a minimum and maximum number of instances and the logic for when to increase or decrease resources.



LALIGA
BEYOND STATS

Powered by Microsoft

15
EN-NESYRI

7
SUSO

 Microsoft
Azure

TOP 3



Questions





Step-1

Field Name	Data Type	Description
AssetID	BigInt, Autoincrementing	An ID for each asset tuple
AssetName	Varchar(255)	A label for the asset
Cost	Integer	The valuation in roubles
AssetType	Varchar(255)	A label for the asset type
NameOfOwner	Varchar(255)	The Name of the current owner
AddressLine1	Varchar(255)	The first line of an address
AddressLine2	Varchar(255)	The second line of its address
Note	Text	A miscellaneous note

Step-2

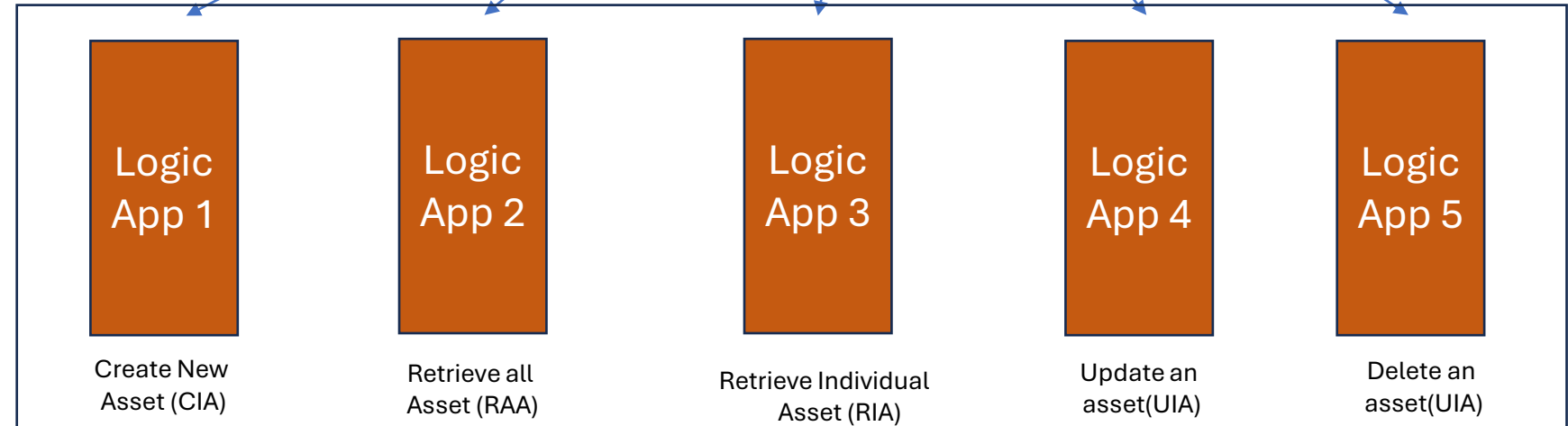


Azure
SQL DB

GET
POST
DELETE
PUT

Week-8 Lab REST API Implementation

Step-3



Step-4 Test

Asset Tracking Web App
Week-9 Lab

Week-6 Lab: Logic Apps

